Unit 56 Instrumental analytical methods

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Reference Website
Refer to the following website for information on modern chemical techniques:
http://cd1.edb.hkedcity.net/cd/science/chemistry/online_resources_c.html
(accessed May 2015)

56.1 The use of instruments in analytical chemistry

We can classify analytical methods according to the property that is observed in the measurement process. Table 56.1 lists some of these properties and examples of methods that are based on them.

Table 56.1 Some properties that are observed in measurement process of analytical chemistry

Property measured	Analytical method(s) based on measurement of the property
Mass	• gravimetric analysis
Volume	• volumetric analysis
Absorption of electromagnetic radiation	• colorimetry • infrared spectroscopy
Mass-to-charge ratio	• mass spectrometry

The gravimetric and volumetric analyses described in Unit 55 require samples of the order of a fraction of a gram to a few grams. In addition, these analyses essentially 'destroy' the sample and the original sample cannot be recovered*.

* If the original sample cannot be recovered, the analysis is called a destructive analysis.

In many cases, only very small samples (e.g. a few milligrams) are available. Often chemists are not allowed to destroy samples during analysis, for example, in the identification of artworks. Fortunately there is a range of instrumental analytical methods that help overcome these difficulties.

Instrumental analytical methods have a number of advantages over the more traditional gravimetric and volumetric analyses. These include:

- speed (therefore allowing for the testing of more samples);
- increased level of accuracy;

- less room for human error;
- sensitivity;
- non-destructiveness.



Fig. 56.1 A chemical instrument (a gas chromatograph)

While instrumental analytical methods have a remarkable impact in modern laboratories, gravimetric and volumetric procedures still have a vital role. They are used:

- in preliminary testing;
- in remote environments;
- where spending on expensive equipment is not justified;
- to create standard solutions and to perform independent analyses so that the instruments can be calibrated.

In this unit, we will consider the following instrumental analytical methods:

- colorimetry;
- infrared spectroscopy; and
- mass spectrometry.

56.2 Electromagnetic radiation

Visible light is a form of **electromagnetic radiation**. Other forms of electromagnetic radiation include radio waves, microwaves, infrared radiation, ultraviolet rays, X-rays, and gamma rays.

Scientists use two models to describe the behaviour of electromagnetic radiation — the **wave model** and the **particle model**.

Electromagnetic radiation can be described as a wave, with a characteristic wavelength and frequency. **Wavelength** (symbol: λ) is the distance between adjacent wave crests (Fig. 56.2). Visible light has wavelengths ranging from 400 – 700 nanometres*.

* 1 nanometre (nm) = 10^{-9} m

electromagnetic radiation 電磁輻射　wave model 波動模型　particle model 粒子模型
wavelength 波長

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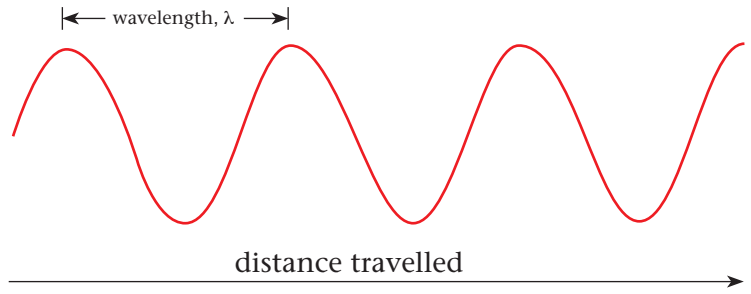


Fig. 56.2 Wavelength is the distance between adjacent wave crests

Frequency (symbol: ν) is the number of complete oscillations that the wave makes each second. The unit of frequency is s^{-1} or hertz (Hz). The relation between frequency and wavelength is

$$\nu\lambda = c$$

where c is the speed of light.

The speed of light has a value of 3.00×10^8 m s^{-1} in a vacuum.

The **electromagnetic spectrum*** (Fig. 56.3) covers various forms of electromagnetic radiation with an enormous range of wavelengths and frequencies, from radio waves at the low-frequency end to gamma rays at the high-frequency end.

* We have discussed the electromagnetic spectrum in Topic 12 Patterns in the Chemical World.

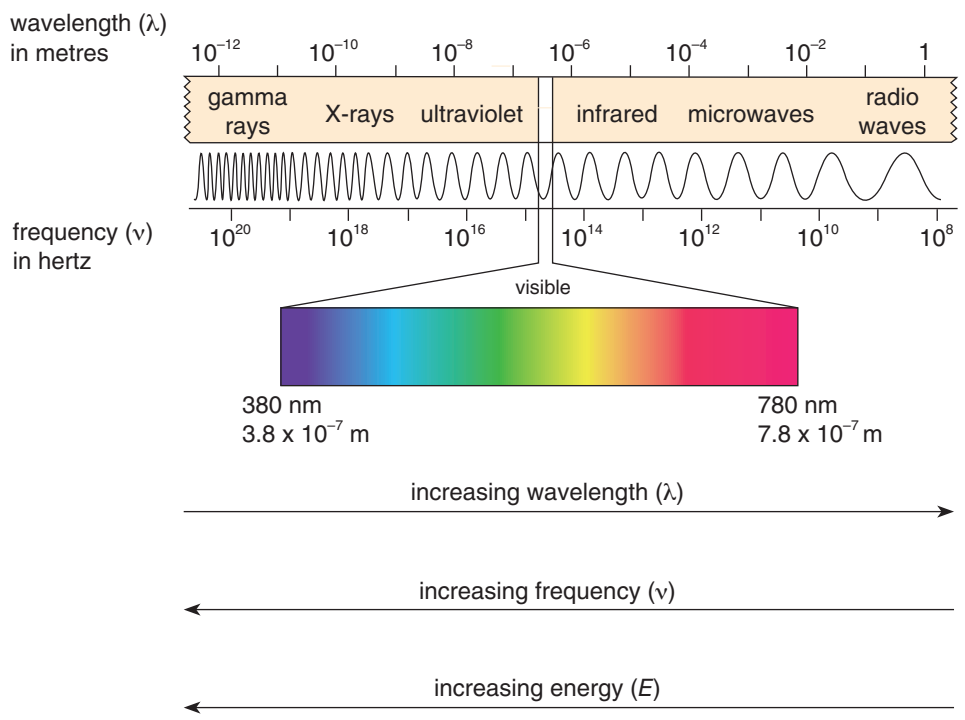


Fig. 56.3 The electromagnetic spectrum

frequency 頻率　　electromagnetic spectrum 電磁光譜

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In many radiation / matter interactions, it is most useful to consider radiation as a stream of tiny 'packets' of energy called **photons**. We can relate the energy of a photon (E) to its wavelength and frequency by:

$$E = h\nu = \frac{hc}{\lambda}$$

where h is **Planck's constant** (= 6.63×10^{-34} J s)* and c the speed of light (= 3.00×10^{8} m s^{-1}).

* The relationship was initially proposed by Max Planck in 1900.

56.3 The interaction of radiation with matter

Electromagnetic radiation can interact with matter, transferring energy to the chemicals involved. The chemicals absorb energy and the absorbed energy can cause changes in the chemicals. Just what changes occur depend on:

- the chemical involved;
- the amount of energy absorbed.

When an atom absorbs radiation, electrons in the atom jump to higher shells.

Look at chemicals with molecular structures. Molecules move around and rotate. The bonds in the molecules also vibrate all the time. Electrons in molecules can move between the different shells.

These different kinds of behaviour involve different amounts of energy — for example, making the bonds in a molecule vibrate generally involves more energy than making the whole molecule rotate. The energy needed increases in the general order as shown in Fig. 56.4.

Notice that vibrational energy of a certain bond in a molecule can only occupy certain fixed energy levels. The vibrational energy of the bond in a HCl molecule moves from one energy level to a higher energy level only when the HCl molecule absorbs energy of radiation* of frequency that matches the frequency of vibration of the bond in the molecule.

* It happens that this radiation corresponds to the infrared region of the electromagnetic spectrum. We will discuss this later in this unit.

photon 光量子　　Planck's constant 普朗克常數

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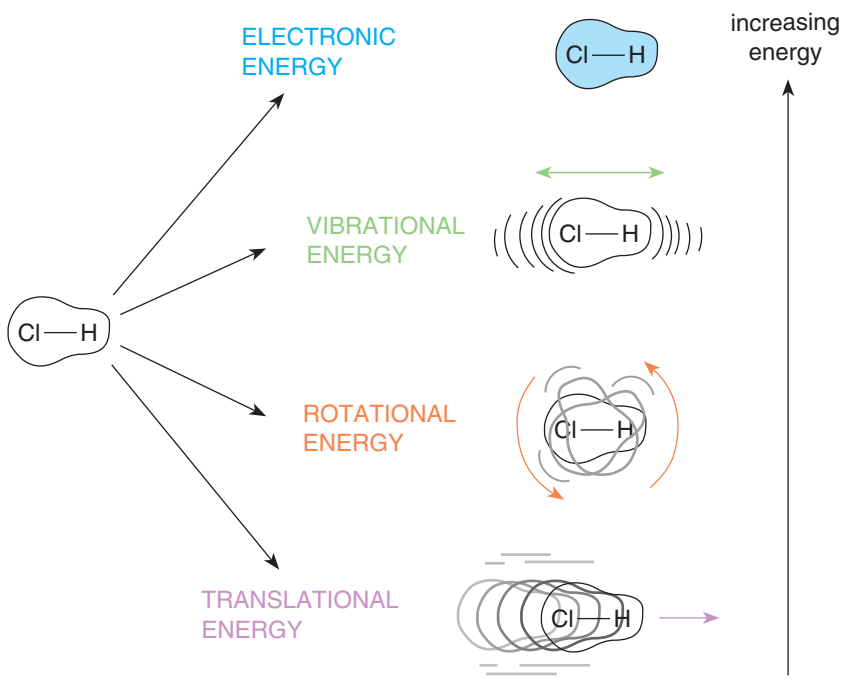


Fig. 56.4 An HCl molecule has energy associated with different aspects of its behaviour

56.4 Colorimetry

When visible light hits a transparent solution, part may be absorbed and part may be transmitted. Visible light can cause electrons jumping to higher shells.

The radiation transmitted and reflected towards our eyes contribute to the colour which we perceive. For example, a blue copper(II) sulphate solution absorbs radiation in the orange region and transmits radiation in the blue region of the visible spectrum* (Fig. 56.5). A N1 concentrated solution of copper(II) sulphate is more strongly coloured than a diluted solution because the orange light is more strongly absorbed by the concentrated solution (Fig. 56.6).

* In Topic 12 Patterns in the Chemical World, we have learnt that in general, the colour that we see is complementary to the colour of the light absorbed.

N1 Refer to page T68.

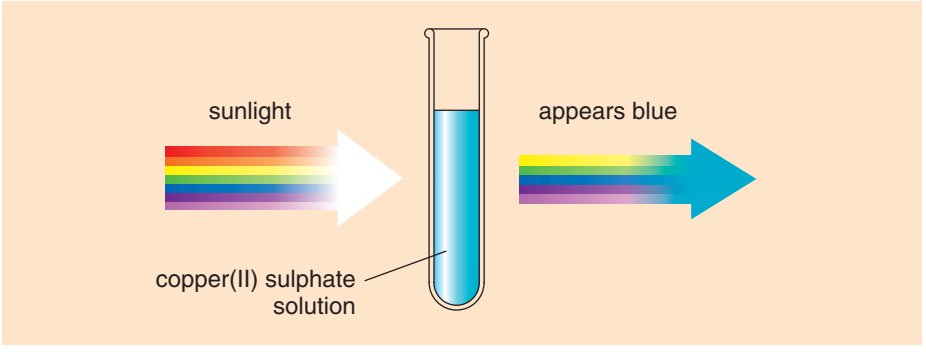


Fig. 56.5 The copper(II) sulphate solution appears blue because radiation, mainly in the blue region of the visible spectrum, is reflected towards our eyes

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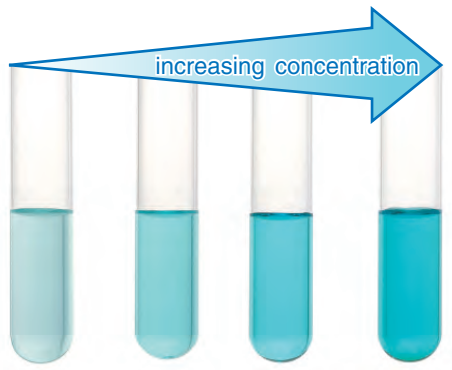


Fig. 56.6 Solutions of copper(II) sulphate of different concentrations

N2 Applications of colorimetry

Colorimetry is used in industries like colour printing, textile manufacturing and paint manufacturing. It is also widely used in food and drug industries. For example, it can be used to determine the concentration of a colouring in a prepared beverage recipe. It can also be used to analyze aspirin.

Any method that determines the concentration of a species based on its ability to absorb certain parts of the radiation of the visible spectrum is known as **colorimetry**. N2

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Components of a colorimeter

A **colorimeter*** is an instrument for measuring the amount of light absorbed by a sample when a beam of light passes through the sample. Fig. 56.7 shows the basic components of a colorimeter.

* We have discussed the colorimeter in Topic 10 Rate of Reaction.

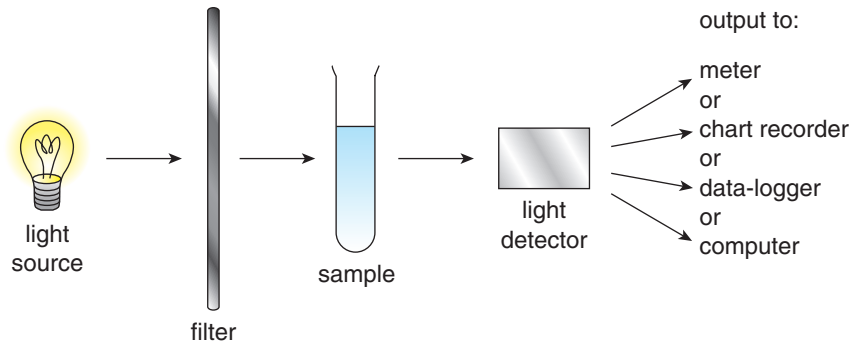


Fig. 56.7 The basic components of a colorimeter

Determining the concentration of the species being analyzed in a sample from a calibration curve

To determine the concentration of a species in a sample solution, we shine the light of wavelength which the species absorbs the most through the sample.

colorimetry 比色法　　colorimeter 比色計

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* We have discussed how to obtain these spectra in Topic 12 Patterns in the Chemical World.

* The orange filter absorbs light in the blue region and allows orange light to pass through.

Suppose we want to determine the concentration of tetraamminecopper(II) ions ($[Cu(NH_3)_4]^{2+}$) in a sample solution. Fig. 56.8 shows the *absorption spectrum** of the ion. The species absorbs most strongly in the orange region of the visible spectrum. Thus, we use an orange filter* in the colorimeter.

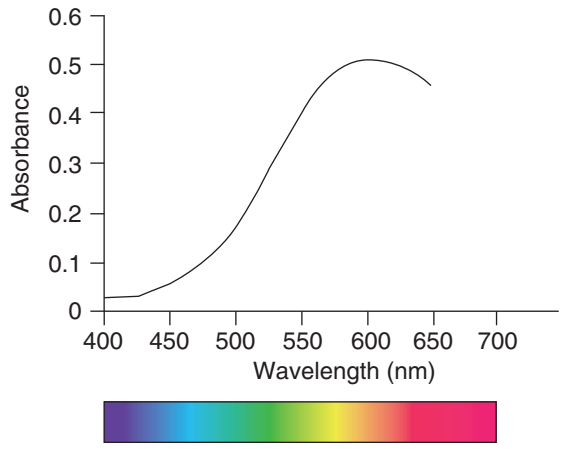


Fig. 56.8 Absorption spectrum of tetraamminecopper(II) ion

Next, prepare a series of standard solutions of tetraamminecopper(II) ions. Using light of the selected colour, record the absorbance of each solution and plot the absorbance against the concentration of tetraamminecopper(II) ions. This is known as the **calibration curve**.

* For very dilute solutions, parts per million (ppm) by mass is a convenient way to express concentration.

$$C_{ppm} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6 \text{ ppm}$$

where C_{ppm} is the concentration in parts per million. Units of mass in the numerator and denominator must agree.

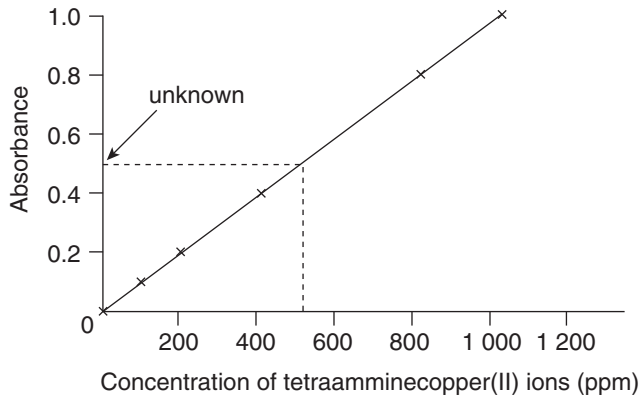


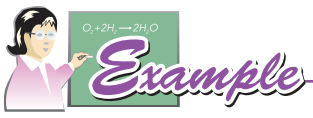
Fig. 56.9 A typical calibration curve for tetraamminecopper(II) ion. The concentrations of standard solutions prepared were 0, 100, 200, 400, 800 and 1 000 ppm* of tetraamminecopper(II) ions

Absorbance is directly proportional to the concentration of the tetraamminecopper(II) ions in a solution. From the calibration curve, we can see that an unknown sample solution with an absorbance of 0.5 must have a concentration of 500 ppm of tetraamminecopper(II) ions.

absorption spectrum 吸收光譜　　calibration curve 校準曲線

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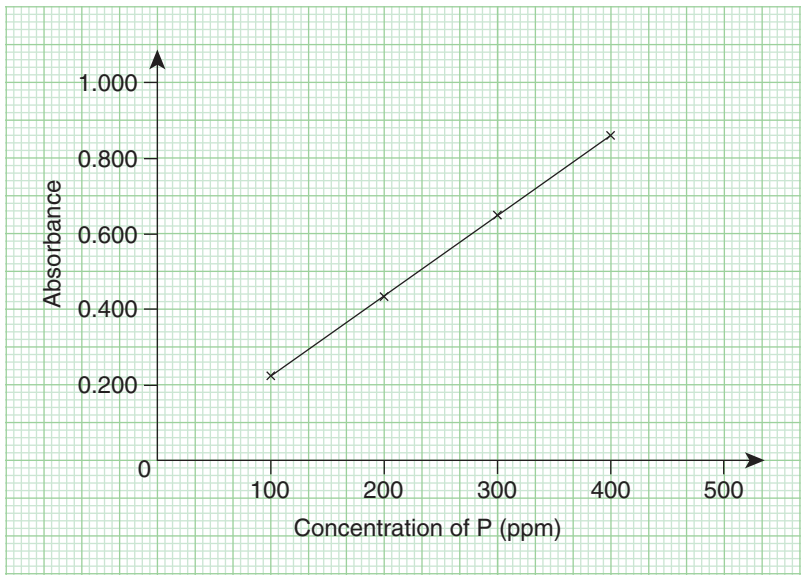
We can use the calibration curve to determine the concentration of tetraamminecopper(II) ions in a sample solution, as long as its concentration falls within the range of concentrations used for constructing the calibration curve. If the sample is too concentrated, we can dilute it by a known factor to bring it into this range.



56.1 Calculating the mass of phosphorus eliminated by a patient per day

One common way to determine the phosphorus in urine is to treat the sample to obtain an intense blue-coloured species called molybdenum blue. The absorbance of molybdenum blue can be measured using a colorimeter.

1.00 cm^3 samples of 4 phosphorus standard solutions were treated to give molybdenum blue. The calibration curve obtained is shown below.



A patient produced 1 200 cm^3 of urine in 24 hours. A 1.00 cm^3 sample was treated to give molybdenum blue. The absorbance was found to be 0.542.

a) Using the calibration curve, determine the concentration of phosphorus in the urine sample.

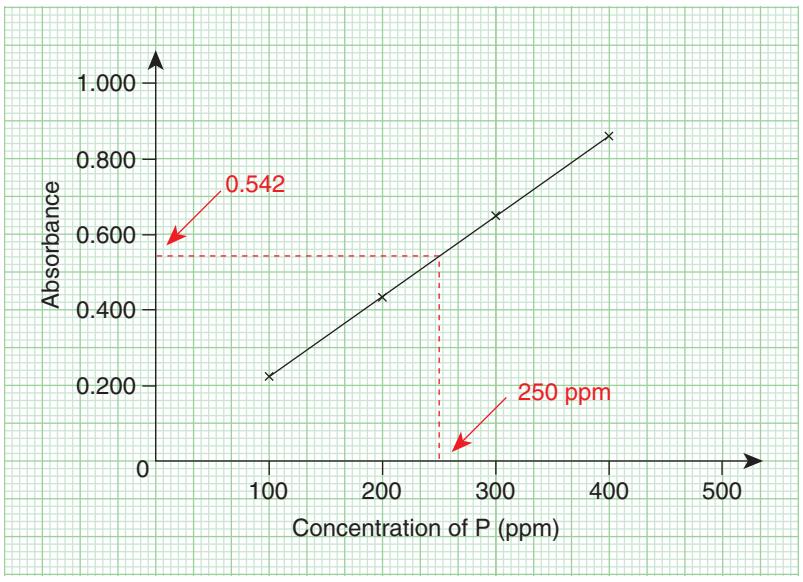
b) What mass of phosphorus was eliminated by the patient per day?

(Relative atomic mass: P = 31.0; density of urine = 1.00 g cm^{-3})

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Solution

a)



250 ppm

b) $C_{ppm} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$ ppm

$250 \text{ ppm} = \frac{\text{mass of P in urine}}{1\ 200 \text{ g}^*} \times 10^6$ ppm

$\text{Mass of P eliminated by the patient per day} = \frac{250 \times 1\ 200}{10^6}$ g

$= 0.300$ g

∴ the mass of phosphorus eliminated by the patient per day was 0.300 g.

* As the density of urine is 1.00 g cm^{-3}, thus the mass of 1 200 cm^3 of urine is 1 200 g.

Checkpoint

Refer to page T68.

One method of determining the concentration of fluoride ions in water uses a red-coloured indicator, In, that reacts with fluoride ions to give a colourless product. The reaction can be represented as follows:

$$\underset{\text{red-coloured indicator}}{In(aq)} + \underset{\text{colourless}}{F^-(aq)} \longrightarrow \underset{\text{colourless}}{FIn^-(aq)}$$

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A calibration curve is prepared in the following way:

- prepare five different standard aqueous solutions of sodium fluoride;
- add X mole of In to 25.00 cm^3 of each of the five standard NaF solutions;
- determine the absorbance of the red In in each mixture using a colorimeter.

The measured absorbances are given in the following table:

Concentration of fluoride ions (mg dm^{-3})		Absorbance of In
Standard solutions	1.00	0.0130
	2.00	0.0110
	3.00	0.0090
	4.00	0.0070
	5.00	0.0050
Unknown NaF sample solution		0.0120

a) Plot the absorbance against the concentration of fluoride ions. Draw a line of best fit.

b) Why does the absorbance fall with increasing concentration of fluoride ions?

c) X mole of In is added to 25.00 cm^3 of a NaF solution of unknown concentration and the absorbance of the mixture determined. The result is shown in the table above. Using the calibration curve drawn in (a), determine the concentration of fluoride ions in this NaF solution in mg dm^{-3}.
1.50 mg dm^{-3}

d) What is the value of X? 9.87×10^{-6}
(Hints: X mol of In is used up at the point where the absorbance reaches zero.)

(Relative atomic mass: F = 19.0)

56.5 Infrared spectroscopy

Infrared (IR) spectroscopy is a common spectroscopic technique used by chemists. Simply, it is the absorption measurement of infrared radiation of different frequencies by a sample. The main goal of infrared spectroscopy is to determine the functional groups present in carbon compounds as different covalent bonds absorb infrared radiation of characteristic frequencies. Thus, infrared spectroscopy is an important and popular tool for structural determination and compound identification. N3

N3 Refer to page T69.

infrared spectroscopy 紅外光譜法

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Reference Website
Refer to the following website for a video clip on infrared spectroscopy:
http://www.youtube.com/watch?v=DDTIJgIh86E
(accessed May 2015)

The energy needed to make vibrational energy changes occur in bonds in molecules corresponds to the infrared region of the electromagnetic spectrum.

If we shine a beam of infrared radiation through a sample, the sample will only absorb radiations of frequencies which match the frequencies of vibration of bonds in it. The rest of the radiation will pass through.

Simple diatomic molecules, such as HCl, HBr and HI, can only vibrate in one way, that is by stretching, where the atoms pull apart and then push together again as if they were joined by a spring (Fig. 56.10).



Fig. 56.10 A stretching vibration in a diatomic molecule

More complex molecules can have very complex vibrations. For example, carbon dioxide can vibrate as shown in Fig. 56.11.

Animation
Vibrations in the CO_2 molecule

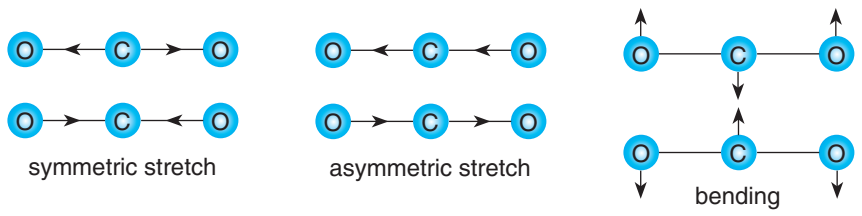


Fig. 56.11 Vibrations in the CO_2 molecule

56.6 Basic features of an infrared spectrum

Fig. 56.12 shows an infrared spectrum of ethanol, CH_3CH_2OH, in the gas phase. This is a plot of percent transmittance (the percentage of infrared radiation which passes through the sample) against wavenumber of the radiation.

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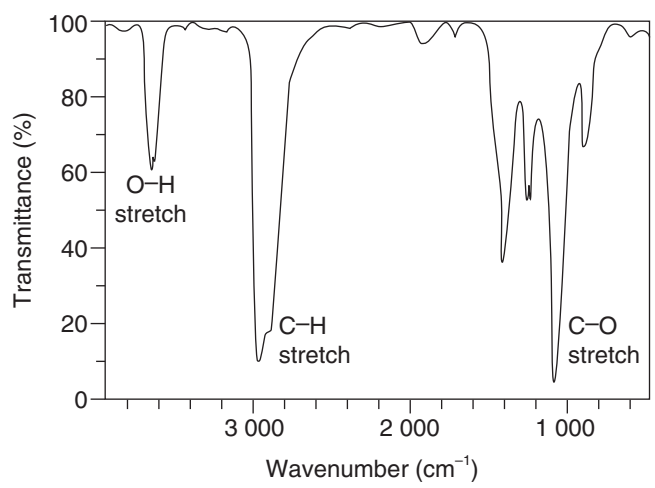


Fig. 56.12 The infrared spectrum of ethanol in the gas phase

It is the **wavenumber** (symbol: $\bar{\nu}$), rather than frequency, which is recorded on an infrared spectrum. The wavenumber is defined as the number of waves per centimetre and has the unit of cm^{-1}. Wavenumbers and wavelengths are related by the following equation:

$$\bar{\nu} \text{ (in cm}^{-1}) = \frac{1}{\lambda \text{ (in cm)}}$$

The wavenumber is directly proportional to the frequency of the radiation ($\bar{\nu} \propto \nu$)*.

Look at Fig. 56.12 again. High values of percent transmittance mean that the sample is absorbing little or no radiation. There are strong dips at around 3 600 cm^{-1}, 3 000 cm^{-1} and 1 050 cm^{-1}. These dips mean that the radiation is being absorbed. They are also called **absorption bands**. We say that these bands are strong because of the large amount of absorption.

Fig. 56.13 shows some of the vibrations in an ethanol molecule which give rise to the bands:

- the band at about 3 600 cm^{-1} results from the stretching of the O–H bond;
- the band at about 3 000 cm^{-1} results from the stretching of the C–H bond; and
- the band at about 1 050 cm^{-1} results from the stretching of the C–O bond.

* As frequency and wavelength are related by the equation $c = \nu\lambda$ (where c is the speed of light, a constant), thus we can write

$$\frac{c}{\lambda} = \nu$$

Rearranging the terms, we have

$$c\bar{\nu} = \nu$$

wavenumber 波數　　absorption band 吸收光帶

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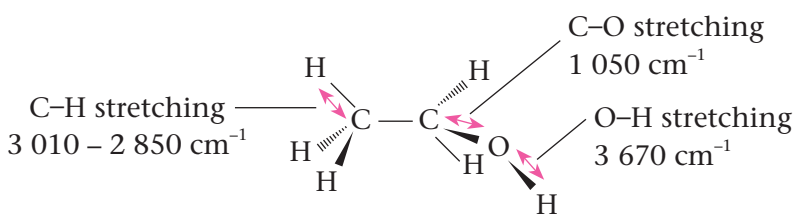


Fig. 56.13 A diagram showing some vibrations in an ethanol molecule which give rise to the absorption bands

56.7 Characteristic absorption wavenumber ranges for different bonds in carbon compounds

In general, we can match a particular bond to a particular infrared absorption region. Fig. 56.14 shows wavenumber ranges at which some covalent bonds absorb infrared radiation.

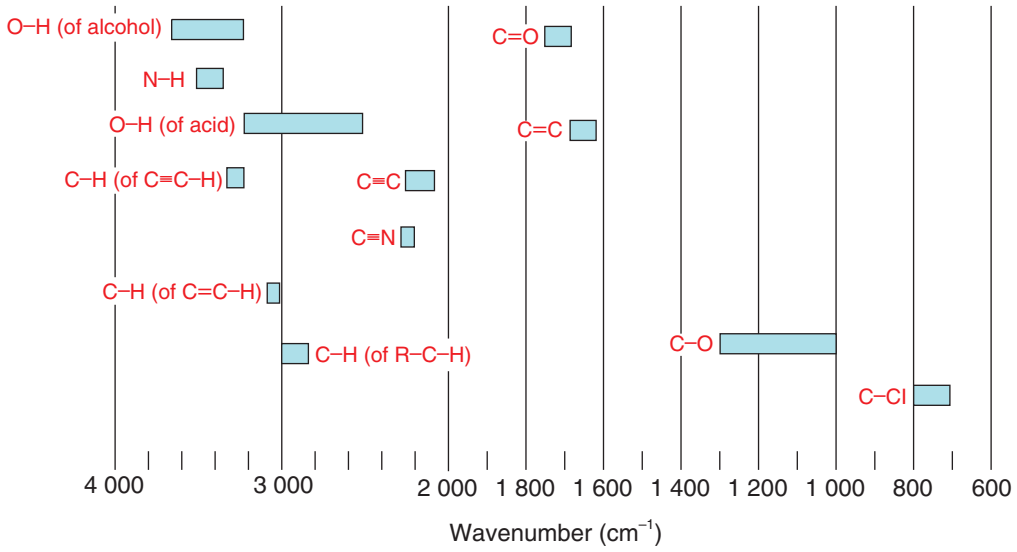


Fig. 56.14 The wavenumber ranges at which some covalent bonds absorb infrared radiation

Animation
Characteristic infrared absorption wavenumber ranges

Notice that hydrogen bonding affects the absorption due to the stretching of the O–H bond. Fig. 56.12 shows the infrared spectrum of ethanol in the gas phase. There is little hydrogen bonding and the O–H absorption is a sharp peak at 3 670 cm^{-1}.

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Now look at Fig. 56.15, which shows the infrared spectrum of a liquid film of ethanol. The absorption band which results from the streching of O–H bond becomes much broader because the O–H vibration is complicated by the hydrogen bonding between hydroxyl groups.

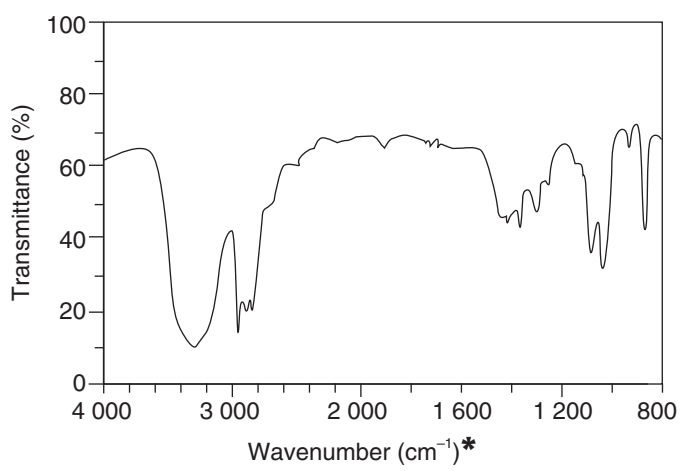


Fig. 56.15 The infrared spectrum of ethanol (liquid film)

* Notice that the spectrum has a non-linear scale for the horizontal axis. This is common for many instruments, so always check the scale carefully when reading off absorption wavenumber values.

* Remember that only those radiations of frequencies which match the frequencies of vibration of bonds will be absorbed.

The frequency of the stretching mode of vibration of a covalent bond is related to the masses of the bonded atoms and the strength of the bond. Light atoms vibrate at higher frequencies than heavier ones. For example, bonds involving the hydrogen atom vibrate with relatively high frequencies and hence absorb infrared radiation of high frequencies* (or wavenumbers) as shown below:

Bond	Absorption wavelength range (cm^{-1})
C–H	2 840 – 3 095
N–H	3 350 – 3 500

Double bonds are stronger and thus absorb radiation of higher frequencies. For example, the C=O bond absorbs radiations of higher frequencies (or wavenumbers) than the C–O bond as shown below:

Bond	Absorption wavelength range (cm^{-1})
C=O	1 680 – 1 750
C–O	1 000 – 1 300

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56.8 Using infrared spectrum in the identification of bonds (or functional groups) in carbon compounds

Since we can match particular bonds (or functional groups) with particular infrared absorption wavenumber ranges, we can identify the presence of particular bonds (or functional groups) in a carbon compound from wavenumbers of absorption bands in the infrared spectrum of the compound.

In general, we can split an infrared spectrum into regions as shown in Table 56.2 for easy interpretation and identification.

Table 56.2 Regions in the infrared spectrum where typical absorptions occur

Absorption range (cm^{-1})	Bonds responsible	Examples
Below 1 500	various	C–O, C–X (halogens)
1 500 – 2 000	double bonds	C=C, C=O
2 000 – 2 500	triple bonds	C≡C, C≡N
2 500 – 4 000	single bonds formed by hydrogen	C–H, O–H, N–H

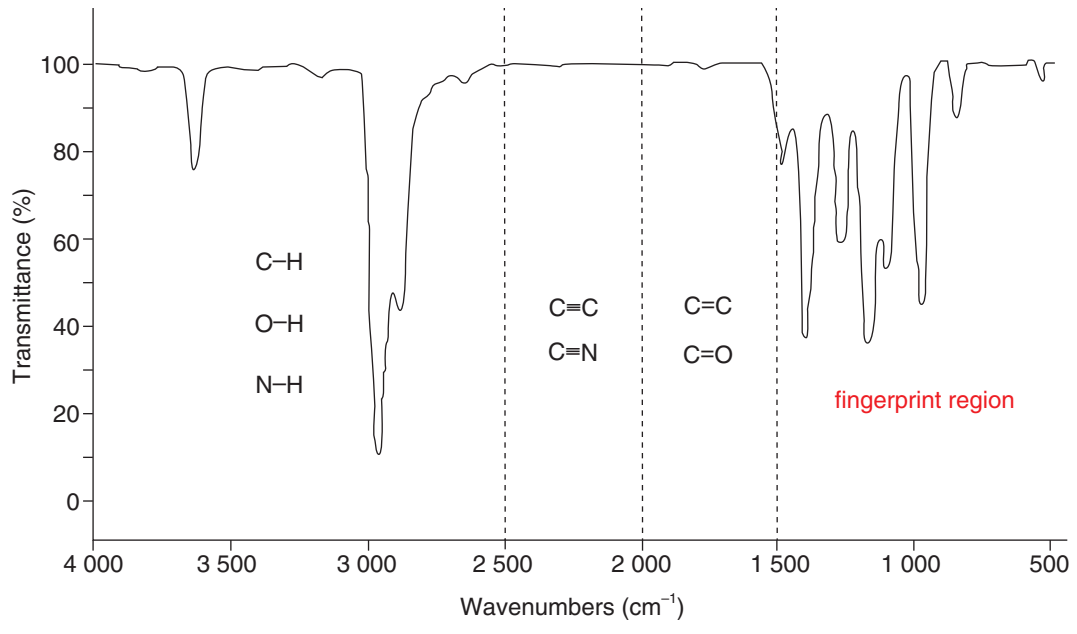


Fig. 56.16 Splitting an infrared spectrum into regions for easy interpretation and identification

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Refer to Fig. 56.16. Below 1 500 cm^{-1} the spectrum can be quite complex and it is more difficult to assign absorptions to particular bonds. This region is characteristic of the particular molecule and is often called the **fingerprint region**. It is useful for identification purposes; for example, we can confirm the identity of a sample by comparing its spectrum with those of known compounds. It is only rarely used to identify the bonds present.

Table 56.3 shows the characteristic infrared absorption wavenumber ranges for some covalent bonds (stretching mode of vibration).

* You need not remember these characteristic wavenumber ranges. This table will be provided in examinations.

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Table 56.3* Characteristic infrared absorption wavenumber ranges for some covalent bonds (stretching mode of vibration)

Bond	Compound type	Wavenumber range (cm^{-1})	Intensity (s = strong, m = medium)
C–Cl	—	700 – 800	s
C–O	alcohols, esters	1 000 – 1 300	s
C=C	alkenes	1 610 – 1 680	m
C=O	aldehydes, ketones, acids, esters	1 680 – 1 750	s
C≡C	alkynes	2 070 – 2 250	m
C≡N	nitriles	2 200 – 2 280	m
O–H	acids (hydrogen-bonded)	2 500 – 3 300	m (broad)
C–H	alkanes, alkenes	2 840 – 3 095	m – s
O–H	alcohols (hydrogen-bonded)	3 230 – 3 670	s (broad)
N–H	amines	3 350 – 3 500	m

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AL 2010 Paper 2 Q5(c)

AL 2006 Paper 2 Q5(c)
AL 2006 Paper 2 Q6(b)(i)

N4 N5 N6

DSE 2013 Paper 2 Q3(c)(iv)

AL 2005 Paper 2 Q7(a)
AL 2006 Paper 2 Q5(c)
AL 2006 Paper 2 Q6(b)(i)
AL 2008 Paper 2 Q6(a)
AL 2009 Paper 2 Q5(c)
AL 2010 Paper 2 Q5(c)
AL 2012 Paper 2 Q9(a)

Notice the following important points when identifying particular bonds (or functional groups) present in a carbon compound using an infrared spectrum:

1 We need not explain the whole spectrum; we look for one or two signals which are characteristics of particular bonds.

2 Focus on absorption bands at or above 1 500 cm^{-1}. Concentrate first on major absorption bands.

3 The absence of absorption bands at certain characteristic wavenumber ranges provides important information as well. We can use the information to deduce that certain bonds (or functional groups) do not exist in the molecule of the sample being analyzed.

N4 Refer to page T70.

N5 Refer to page T71.

N6 Refer to page T71.

fingerprint region 指紋區

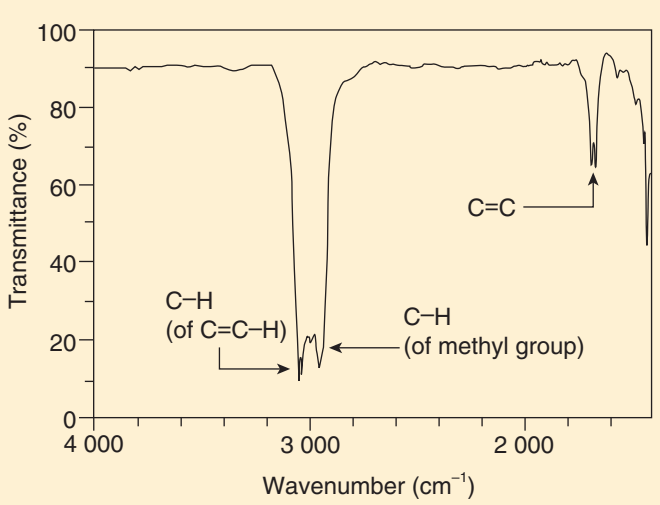
Unit 56(E).indd 169 2015/6/30 11:41:16 AM

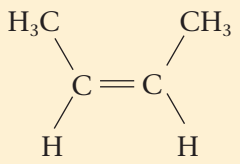
56.9 Interpreting infrared spectra

Let us now apply the data given in Table 56.3 to interpret the infrared spectra of carbon compounds.

Cis-but-2-ene

The spectrum of cis-but-2-ene (Fig. 56.17) shows absorptions at 1 657 cm^{-1} and 1 677 cm^{-1}, indicating the presence of a C=C bond. The spectrum shows two types of C–H absorptions: just below 3 000 cm^{-1} for the C–H bond of the methyl group and just above 3 000 cm^{-1} for the C–H bond of C=C–H.





Wavenumber range (cm^{-1})	Bond
1 657 1 677	C=C
2 952	C–H (of methyl group)
3 044	C–H (of C=C–H)

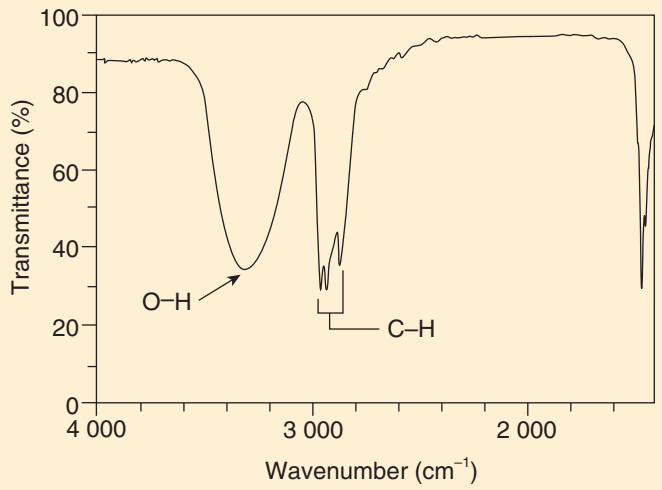
Fig. 56.17 Infrared spectrum of *cis*-but-2-ene

Butan-1-ol

The spectrum of butan-1-ol (Fig. 56.18) shows a broad absorption band at 3 330 cm^{-1} which is characteristic of an O–H bond (hydrogen-bonded).

AL 2001 Paper 2 Q7(b)(ii)
AL 2002 Paper 1 Q5(a)
AL 2005 Paper 2 Q5(c)(ii)(III)

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$CH_3CH_2CH_2CH_2OH$

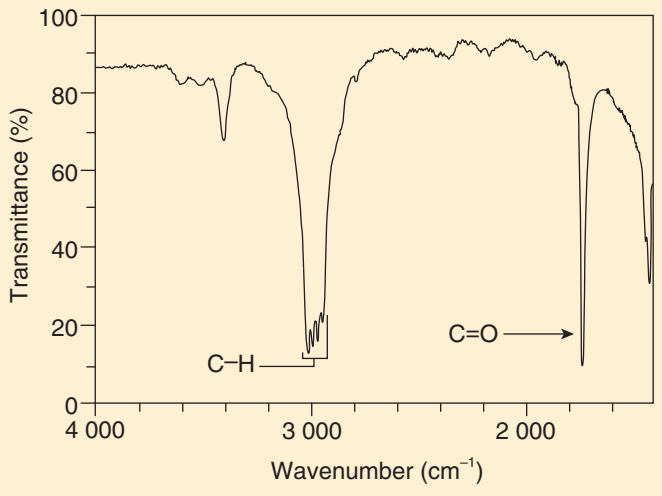
Wavenumber range (cm^{-1})	Bond
2 875 – 2 960	C–H
3 330 (broad)	O–H (hydrogen-bonded)

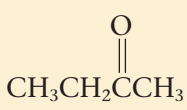
Fig. 56.18 Infrared spectrum of butan-1-ol

AL 2001 Paper 1 Q8(b)(iii)
AL 2001 Paper 2 Q7(b)(ii)
AL 2002 Paper 1 Q5(a)
AL 2003 Paper 2 Q7(c)

Butanone

The spectrum of butanone (Fig. 56.19) shows a strong absorption at 1 720 cm^{-1}, indicating the presence of a C=O bond.





Wavenumber range (cm^{-1})	Bond
1 720	C=O
2 925 – 2 983	C–H

Fig. 56.19 Infrared spectrum of butanone

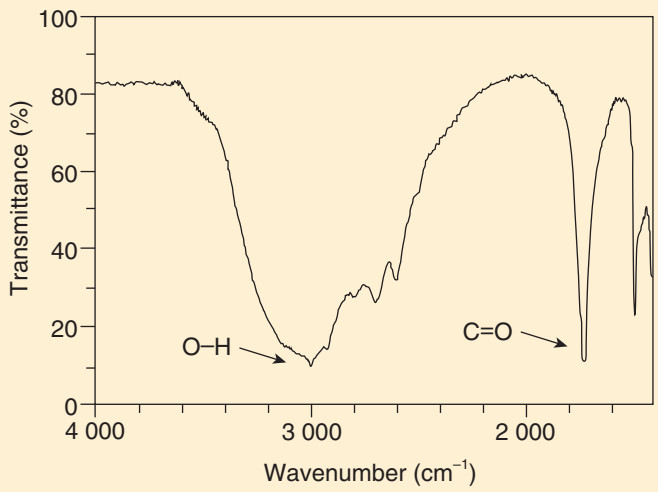
Butanoic acid

The spectrum of butanoic acid (Fig. 56.20) shows a strong absorption at 1 708 cm^{-1}, indicating the presence of a C=O bond. The broad absorption band at 3 100 cm^{-1} is characteristic of an O–H bond (hydrogen-bonded).

AL 2005 Paper 2 Q7(a)

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Notice that the absorption band results from the O–H bond in alcohols and carboxylic acids usually appears as a broad band because the O–H vibration is complicated by the hydrogen bonding between the hydroxyl groups.



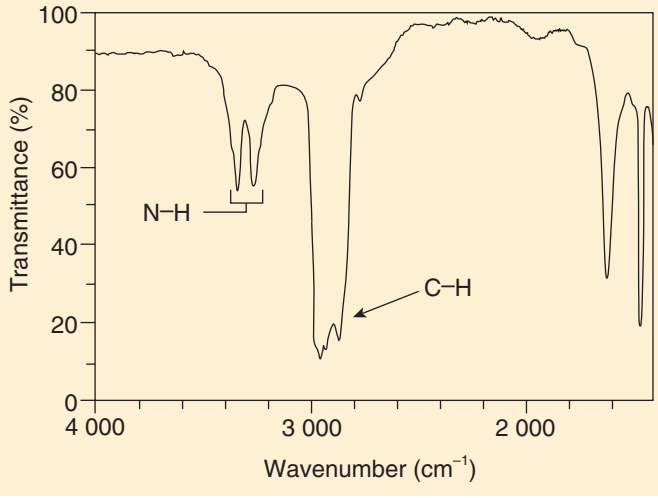
$CH_3CH_2CH_2COOH$

Wavenumber range (cm^{-1})	Bond
1 708	C=O
3 100 (broad)	O–H (hydrogen-bonded)

Fig. 56.20 Infrared spectrum of butanoic acid

Butan-1-amine

The spectrum of butan-1-amine (Fig. 56.21) shows absorptions at 3 280 cm^{-1} and 3 371 cm^{-1}, indicating the presence of an N–H bond.



$CH_3CH_2CH_2CH_2NH_2$

Wavenumber range (cm^{-1})	Bond
2 875 – 2 960	C–H
3 280 3 371	N–H

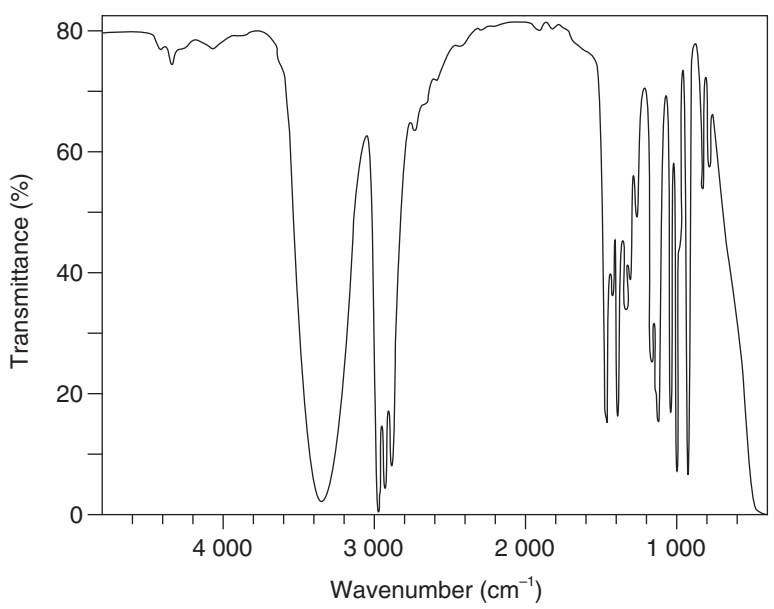
Fig. 56.21 Infrared spectrum of butan-1-amine

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Example

56.2 Deducing whether a compound is butan-2-ol or butanone

Compound A is known to be either butan-2-ol or butanone. The infrared spectrum of A is shown below.



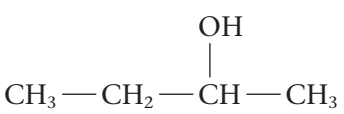
a) Draw the structural formulae of butan-2-ol and butanone.

b) Identify key bands in the spectrum and the bond to which each corresponds. Give the answer in the form of a table.

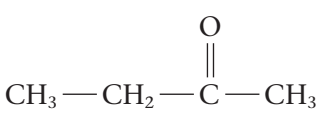
c) Decide whether A is butan-2-ol or butanone.

Solution

a)



butan-2-ol



butanone

b)

Absorption (cm^{-1})	Bond
2 970	C–H
3 360	O–H

c) The infrared spectrum of A shows a broad absorption at about 3 400 cm^{-1}, which is characteristic of an O–H bond (hydrogen-bonded). There is no absorption at about 1 700 cm^{-1}, indicating that A does not contain a C=O bond. Hence A is butan-2-ol.

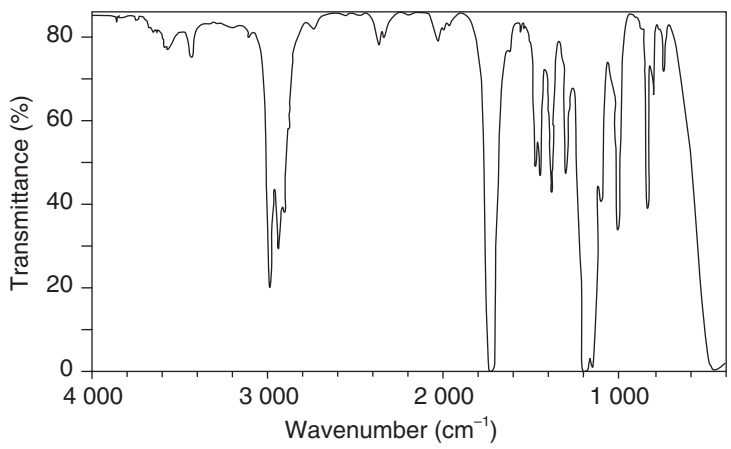
Unit 56(E).indd 173 2015/6/30 11:41:21 AM

56.3 Deducing the structure of a compound

Compound X, with a relative molecular mass of 74.0, has the following composition by mass:

C 48.7% H 8.10% O 43.2%

X gives a negative result when treated with 2,4-dinitrophenylhydrazine. The infrared spectrum of X is shown below.



a) i) Calculate the empirical formula of X.

ii) Calculate the molecular formula of X.

b) Deduce TWO possible structures of X.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

Solution

a) i) Suppose we have 100 g of compound X, so there are 48.7 g of carbon, 8.10 g of hydrogen and 43.2 g of oxygen.

	Carbon	Hydrogen	Oxygen
Mass of element in the compound	48.7 g	8.10 g	43.2 g
Number of moles of atoms that combine	$\frac{48.7\text{ g}}{12.0\text{ g mol}^{-1}} = 4.06\text{ mol}$	$\frac{8.10\text{ g}}{1.0\text{ g mol}^{-1}} = 8.10\text{ mol}$	$\frac{43.2\text{ g}}{16.0\text{ g mol}^{-1}} = 2.70\text{ mol}$
Simplest ratio of atoms	$\frac{4.06\text{ mol}}{2.70\text{ mol}} = 1.50$	$\frac{8.10\text{ mol}}{2.70\text{ mol}} = 3.00$	$\frac{2.70\text{ mol}}{2.70\text{ mol}} = 1.00$
Simplest whole number ratio of atoms	1.50 x 2 = 3.00	3.00 x 2 = 6.00	1.00 x 2 = 2.00

$\therefore$ the empirical formula of X is $C_3H_6O_2$.

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ii) Let $(C_3H_6O_2)_n$ be the molecular formula of X.

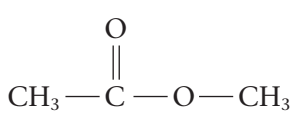
Relative molecular mass of X = n(3 x 12.0 + 6 x 1.0 + 2 x 16.0)
= 74n

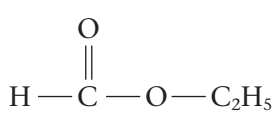
i.e. 74n = 74.0
n = 1

∴ the molecular formula of X is $C_3H_6O_2$.

b)
- X is neither an aldehyde nor a ketone as it gives a negative result when treated with 2,4-dinitrophenylhydrazine.
- The infrared absorption at about 1 200 cm^{-1} indicates the presence of a C–O group.
- The infrared absorption at about 1 720 cm^{-1} indicates the presence of a C=O group.
- The absence of a broad absorption band at 2 500 cm^{-1} – 3 300 cm^{-1} indicates that X is not an alcohol or a carboxylic acid.

X is probably an ester. The possible structures of X are:







Checkpoint

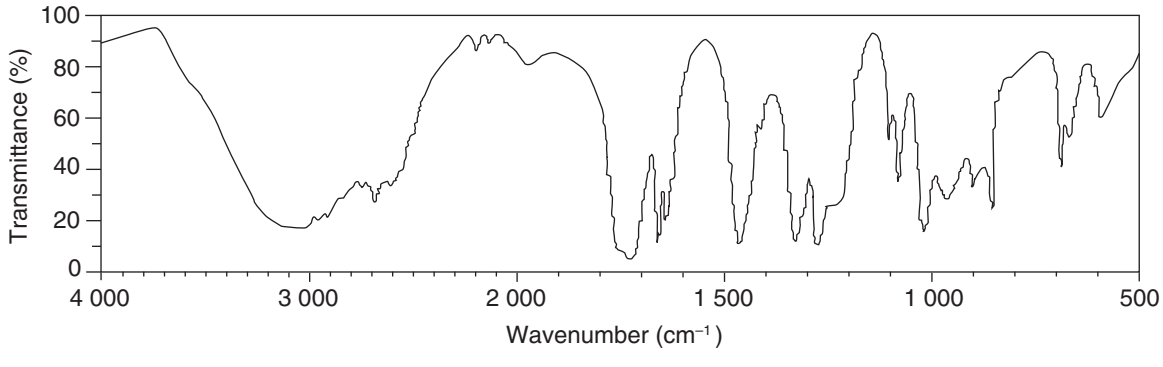
Refer to page T72.

Refer to the data given in Table 56.3 when identifying particular bonds (or functional groups) in compounds.

1 Acrolein (CH_2=CHCHO) and acrylic acid (CH_2=CHCOOH) are used in industry for the manufacture of plastics. Both acrolein and acrylic acid can be made from prop-2-en-1-ol (CH_2=$CHCH_2OH$).

a) Draw the structural formulae of prop-2-en-1-ol and acrolein.

b) Prop-2-en-1-ol can be oxidized to form either acrolein or acrylic acid. A sample of prop-2-en-1-ol was oxidized and an infrared spectrum of the product was obtained.



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Decide whether acrolein (CH_2=CHCHO) or acrylic acid (CH_2=CHCOOH) was formed. Explain your reasoning.

2 Compounds X and Y are isomers. Suggest a possible structure for each compound, X and Y, on the basis of the following information.

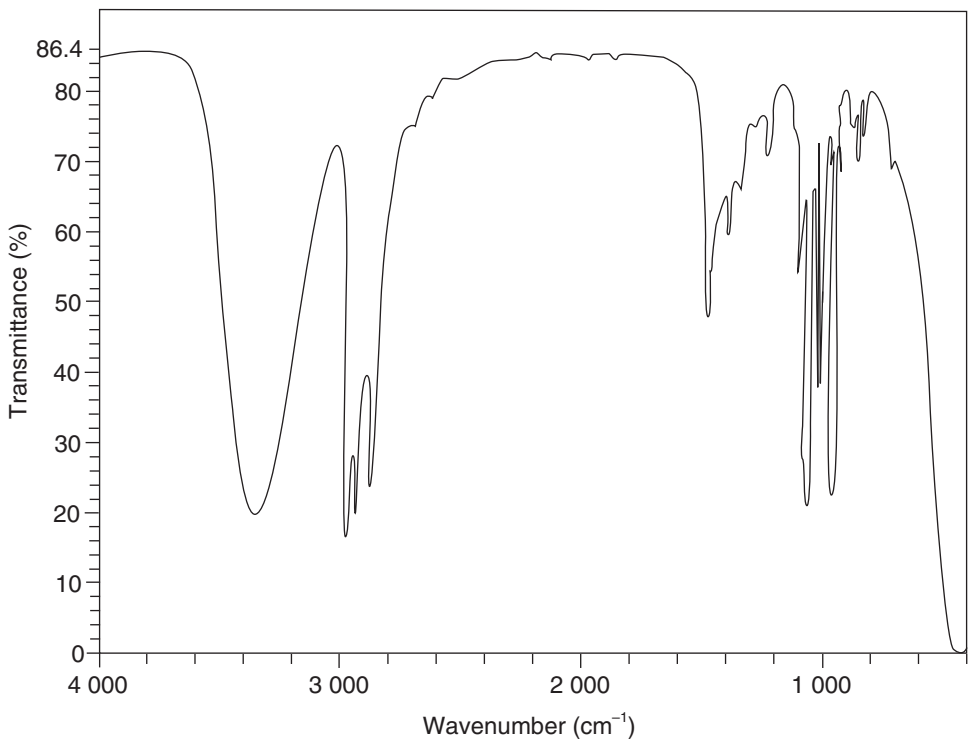
- Analysis data show that they have the following composition by mass: C = 40.0%, H = 6.70%, O = 53.3%.
- Their relative molecular mass is 60.0.
- Each has an absorption in its infrared spectrum at about 1 700 cm^{-1} but only Y has a broad absorption at 3 350 cm^{-1}.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

3 A ($C_6H_{12}O_2$) is a neutral compound and is formed by the reaction between compounds X and Y in the presence of a small amount of concentrated sulphuric acid.

X and Y can both be formed from propanal by different redox reactions.

The infrared spectrum of Y is shown below.



a) Deduce the structural formulae of A, X and Y. Explain your reasoning.

b) Give suitable reagents, in each case, for the formation of X and Y from propanal.

c) State the role of concentrated sulphuric acid in the formation of A.

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56.10 Mass spectrometry

Mass spectrometry is a form of analysis that separates and identifies substances on the basis of mass-to-charge ratios of positive ions formed from substances when they are bombarded by high energy electrons in a vacuum.

The instrument used in mass spectrometry is called a **mass spectrometer**.

What is a mass spectrometer?

Fig. 56.22 shows a mass spectrometer. There are five essential parts of the instrument:

- a sample inlet;
- an ionization chamber;
- an electric field to accelerate a beam of positive ions;
- a magnetic field for deflecting the ion beam;
- an ion detector.

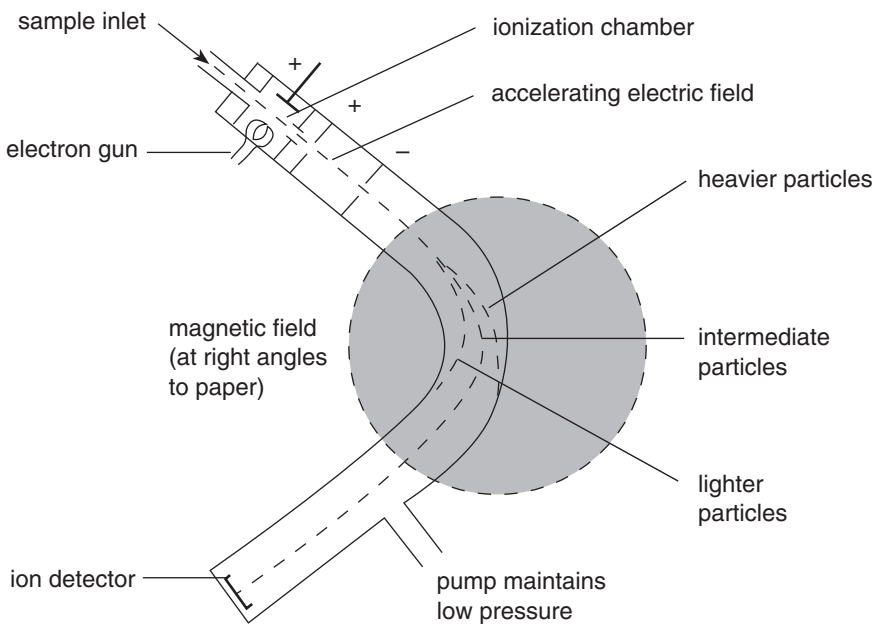


Fig. 56.22 A mass spectrometer

Reference Website
Refer to the following website for a video clip on mass spectrometry:
http://www.youtube.com/watch?v=J-wao0O0_qM&feature=related
(accessed May 2015)

Animation
How a mass spectrometer works

mass spectrometry 質譜法　　mass spectrometer 質譜儀

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A very low pressure must be maintained in the mass spectrometer so that ions can pass through the instrument without colliding with air molecules.

Sample inlet

A sample being analyzed is injected into the chamber and heated until it changes to gaseous form. The sample being analyzed can be an element or a compound.

Ionization chamber

Here the sample is bombarded by a stream of high energy electrons. This knocks off electrons from the sample (may be atoms or molecules), producing positive ions.

$$X(g) + \underset{\text{bombarding electron}}{e^-} \longrightarrow X^+(g) + 2e^-$$

Accelerating electric field

The positive ions are accelerated here. They move towards the magnetic field at a very high speed.

Magnetic field

The positive ions are deflected here. At a given strength of the accelerating electric field and the magnetic field, only ions of one particular mass/charge ratio (m/e) will hit the ion detector at the end of the spectrometer.

By varying the accelerating electric field or the deflecting magnetic field, we can focus each positive ion beam on the detector in sequential order of mass/charge ratio.

Ion detector

The detector produces an electric current when hit by ions. The detector signal is recorded as the magnetic field is gradually changed. This produces a mass spectrum which shows the mass/charge ratio of ions and the corresponding current intensities (i.e. the relative abundance of the ions).

AL 2001 Paper 2 Q1(b)

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N7 Refer to page T73.

Uses of mass spectrometry N7

Uses of mass spectrometry include:

- measuring relative atomic masses and relative molecular masses accurately;
- detecting the isotopes of elements and measuring the relative abundance of the isotopes;
- identifying compounds by comparison with standard spectra;
- determining structural information (from **fragmentation patterns**) of new compounds.

We will consider these uses in the following parts.

56.11 Finding relative atomic masses and relative molecular masses from mass spectra

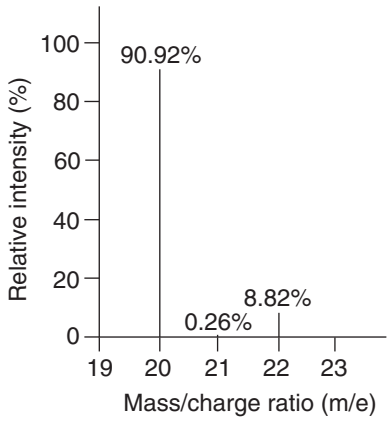


Fig. 56.23 The mass spectrum of naturally occurring neon

Fig. 56.23 shows the mass spectrum of neon. The position of a peak gives the mass/charge ratio (m/e) of an ion and the height of the peak gives the relative abundance of the ion.

On the mass spectra in this book, the x-axis is labelled: mass/charge ratio (m/e). Most of the ions formed carry one positive charge. This means that the charge e = +1, which makes the mass/charge ratios of the ions equal to their relative masses.

AL 2001 Paper 2 Q1(b)

Finding relative atomic masses

In the mass spectrum of an element, different peaks correspond to various isotopes of the element. Thus, we can use the mass spectrum to find the relative masses of isotopes and their relative abundance in a sample of an element. We can work out the relative atomic mass of the element from these information*.

* We have performed such calculations in Topic 2 Microscopic World I.

Finding relative molecular masses

Mass spectra are more commonly used to find relative molecular masses. Fig. 56.24 shows the mass spectrum of butane (C_4H_{10}). N8

N8 Mass spectrometry is often used to determine relative molecular masses because
- relative molecular masses can be determined to very high accuracy;
- only a small amount of the sample is required;
- the time for the experiment is short.

fragmentation pattern 碎裂模式

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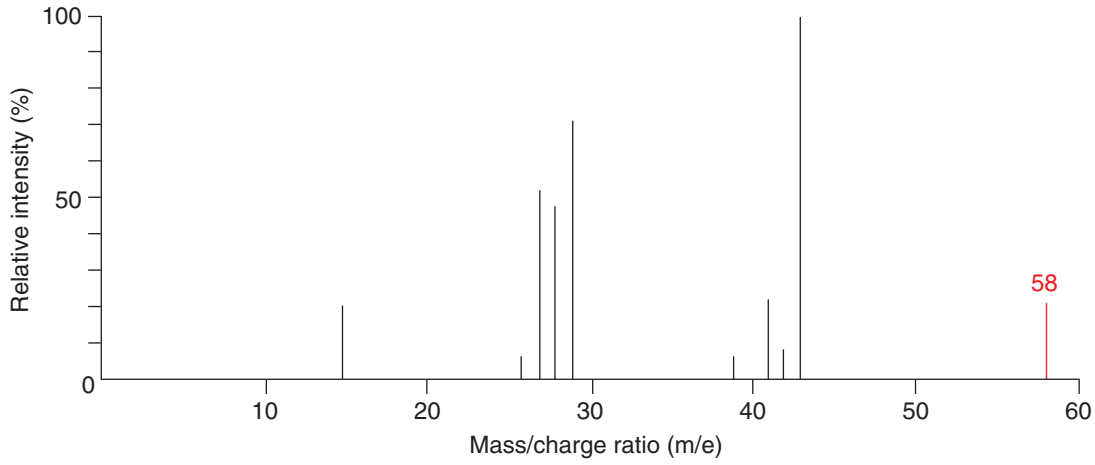


Fig. 56.24 The mass spectrum of butane

In the mass spectrum of a compound, the peak with the highest m/e will most likely correspond to the **molecular ion** ($M^{+\cdot}$).

When one of the electrons of a butane molecule is knocked off by a bombarding electron, a positively charged molecular ion is produced (Fig. 56.25).

$$\underset{\text{molecule}}{CH_3CH_2CH_2CH_3(g)} + \underset{\substack{\text{bombarding}\\ \text{electron}}}{e^-} \longrightarrow \underset{\text{molecular ion}}{[CH_3CH_2CH_2CH_3]^{+\cdot}(g)} + 2e^-$$

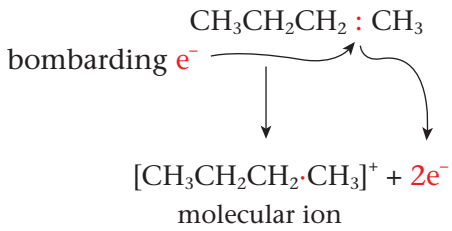


Fig. 56.25 When one of the electrons of a butane molecule is knocked off by a bombarding electron, a positively charged molecular ion is produced

The relative molecular mass of butane is 58.0. In the mass spectrum of butane, the peak at m/e = 58 corresponds to the molecular ion, $[CH_3CH_2CH_2CH_3]^{+\cdot}$. Thus, it is called the **molecular ion peak**.

The other peaks in the mass spectrum are caused by butane molecular ions that have broken down into fragments. Such fragmentation is important in working out the identity of a molecule.

molecular ion 分子離子　　molecular ion peak 分子離子峰

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56.12 Using fragmentation patterns to determine the structures of molecules of carbon compounds

Here we will look at the use of mass spectrometry in determining the structures of molecules. The technique is particularly useful for molecules of carbon compounds.

Refer to the mass spectrum of butane shown in Fig. 56.24 again. Besides the peak at m/e = 58, there are several other peaks because the butane molecular ion fragments in the mass spectrometer.

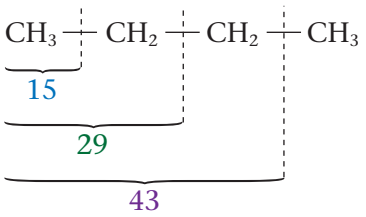
Fragmentation of the molecular ion gives smaller positively charged ions, such as $CH_3CH_2CH_2^+$, $CH_3CH_2^+$ and CH_3^+.

$[CH_3CH_2CH_2CH_3]^{+\cdot}$ →
- $CH_3CH_2CH_2^+ + \cdot CH_3$ (propyl ion, m/e = 43)
- $CH_3CH_2^+ + \cdot CH_2CH_3$ (ethyl ion, m/e = 29)
- $CH_3^+ + \cdot CH_2CH_2CH_3$ (methyl ion, m/e = 15)

Peaks corresponding to these ions are marked on the mass spectrum of butane (Fig. 56.26). Other peaks in the spectrum result from the loss of hydrogen atoms from these fragments.

The most abundant ion gives the strongest detector signal which is set to 100% in the spectrum. This is referred to as the **base peak**. This peak corresponds to the most stable species. The intensities of all the other peaks are expressed as a percentage of its value.

* The uncharged free radical will not produce a line on the mass spectrum. Only charged particles will be accelerated, deflected and detected by the mass spectrometer.



m/e	Ion
58	$[CH_3CH_2CH_2CH_3]^{+\cdot}$ molecular ion peak
43	$CH_3CH_2CH_2^+$ propyl ion peak
29	$CH_3CH_2^+$ ethyl ion peak
15	CH_3^+ methyl ion peak

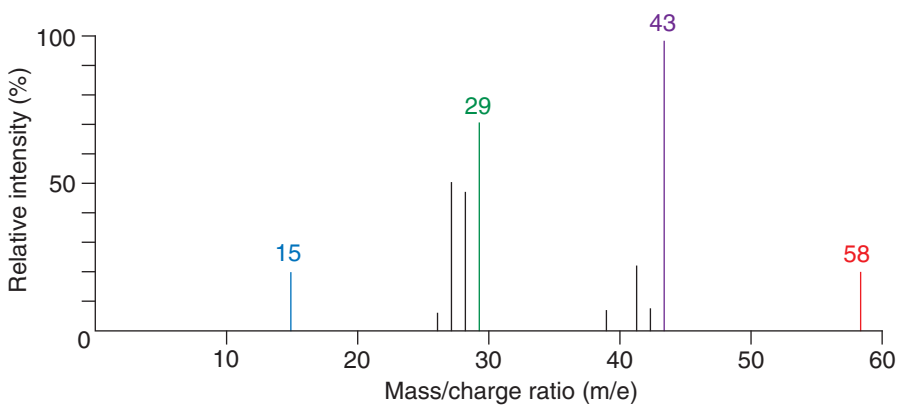


Fig. 56.26 The mass spectrum of butane*, with information on ions corresponding to major peaks

base peak 基本峰

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56.13 Fragmentation patterns of alkanes

Mass spectrum of pentane

Simple straight chain alkane molecules tend to undergo fragmentation by the loss of a free radical. Fig. 56.27 shows the simplified mass spectrum of a straight chain alkane, pentane (relative molecular mass = 72.0).

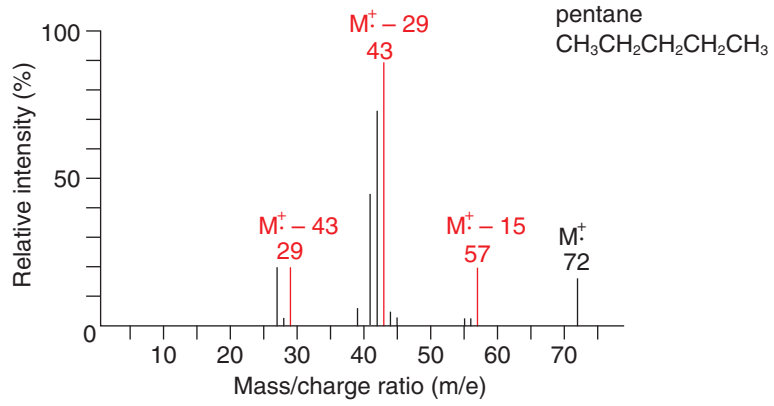


Fig. 56.27 The simplified mass spectrum of a straight chain alkane, pentane

Notice the following peaks in the spectrum:

- the molecular ion peak at m/e = 72;
- a smaller peak at m/e = 57 ($M^{+\cdot} - 15$) corresponding to the loss of a methyl radical ($\cdot CH_3$) from the molecular ion;
- the base peak at m/e = 43 ($M^{+\cdot} - 29$) corresponding to the loss of an ethyl radical ($\cdot CH_2CH_3$) from the molecular ion;
- a smaller peak at m/e = 29 ($M^{+\cdot} - 43$) corresponding to the loss of a propyl radical ($\cdot CH_2CH_2CH_3$) from the molecular ion.

The important fragmentations are as follows:

$$[CH_3CH_2CH_2CH_2CH_3]^{+\cdot} \longrightarrow \begin{cases} CH_3CH_2CH_2CH_2^+ + \cdot CH_3 \\ \text{m/e} = 57 \end{cases}$$

$$[CH_3CH_2CH_2CH_2CH_3]^{+\cdot} \longrightarrow \begin{cases} CH_3CH_2CH_2^+ + \cdot CH_2CH_3 \\ \text{m/e} = 43 \end{cases}$$

$$[CH_3CH_2CH_2CH_2CH_3]^{+\cdot} \longrightarrow \begin{cases} CH_3CH_2^+ + \cdot CH_2CH_2CH_3 \\ \text{m/e} = 29 \end{cases}$$

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Mass spectrum of 2-methylbutane

* A carbocation is an ion with a positively charged carbon.

Chain branching in alkanes increases the likelihood of splitting at a branch point because a more stable carbocation* results. Let us look at the mass spectrum of 2-methylbutane (Fig. 56.28). Notice that 2-methylbutane is an isomer of pentane.

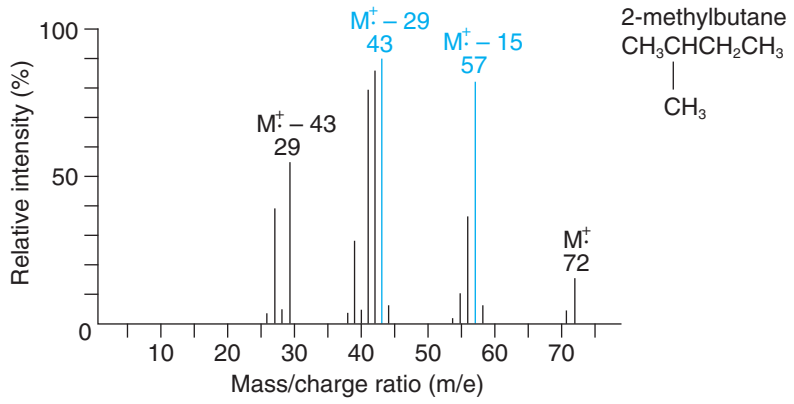
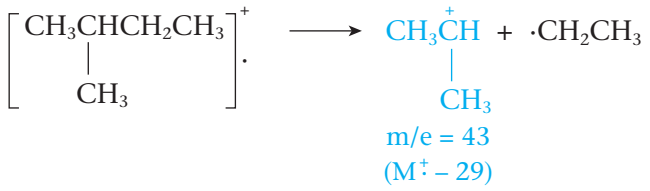


Fig. 56.28 The mass spectrum of 2-methylbutane

Look first at the base peak at m/e = 43. This is caused by a different ion than the corresponding peak in the mass spectrum of pentane. This peak in the spectrum of 2-methylbutane is caused by the following fragmentation:

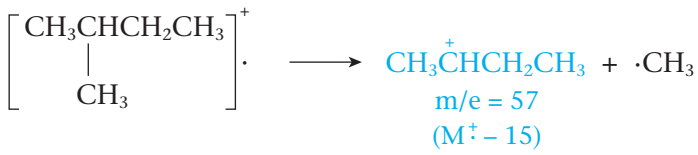


* Order of stability of carbocations: primary < secondary < tertiary

The ion formed is a secondary carbocation* which is relatively stable.

* Notice that the molecular ion of pentane forms a primary carbocation upon the loss of a methyl radical.

The peak at m/e = 57 in the spectrum of 2-methylbutane (Fig. 56.28) is much taller than the corresponding line in the spectrum of pentane (Fig. 56.27). Again a secondary carbocation is formed* — this time, by the following fragmentation:



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Mass spectrum of 2,2-dimethylpropane

Fig. 56.29 shows the mass spectrum of 2,2-dimethylpropane. 2,2-dimethylpropane is an isomer of pentane as well.

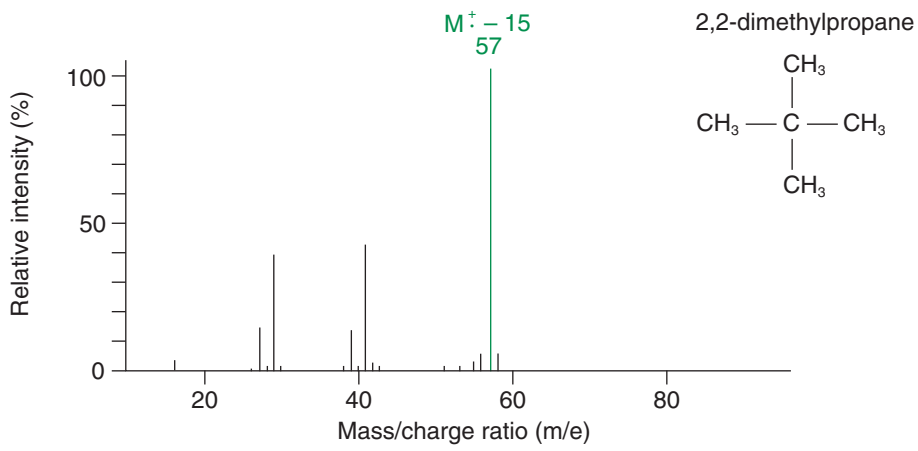
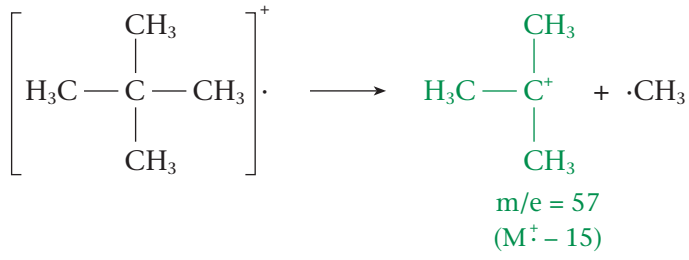


Fig. 56.29 Mass spectrum of 2,2-dimethylpropane

Loss of a methyl radical by the molecular ion produces a tertiary carbocation which is very stable.



This tertiary carbocation ion gives rise to the base peak at m/e = 57. This fragmentation takes place so readily that virtually none of the molecular ions (m/e = 72) exist long enough to be detected.

56.14 Fragmentation patterns of aromatic compounds

For aromatic compounds, their molecular ions tend to fragment so as to yield the benzylic cation, $C_6H_5CH_2^+$, which rearranges to give a relatively stable carbocation $C_7H_7^+$. This gives rise to a characteristic peak at m/e = 91.

DSE 2012 Paper 2 Q3(a)(iv)

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Fig. 56.30 shows the mass spectrum of methylbenzene.

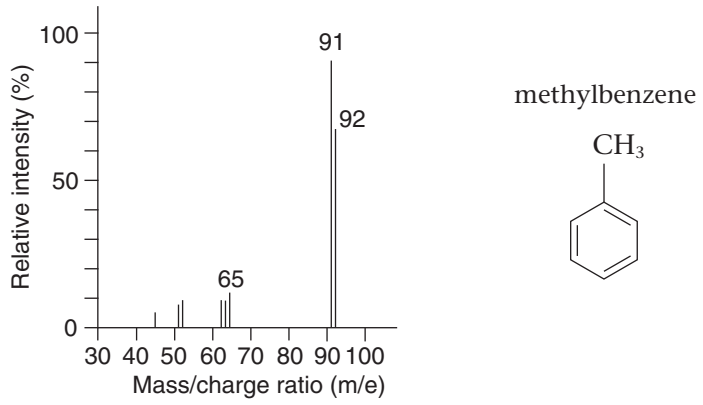
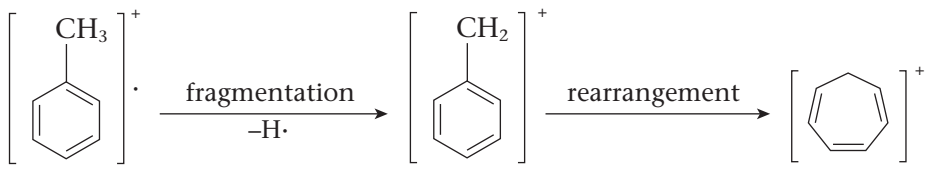


Fig. 56.30 The mass spectrum of methylbenzene

AL 2010 Paper 2 Q5(c)

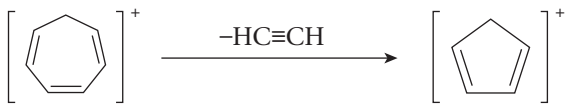
The molecular ion of methylbenzene undergoes fragmentation as follows:



N9 Examination questions often give the mass spectra of aromatic compounds and ask about possible structures of the compounds. Remind students to pay special attention to the following peaks:

- m/e = 77 due to $C_6H_5^+$ ion; and
- m/e = 91 due to $C_6H_5CH_2^+$ ion.

The carbocation $C_7H_7^+$ is usually associated with the fragment $C_5H_5^+$ ion formed by losing a neutral ethyne molecule. This gives rise to a peak at m/e = 65.



m/e = 91 m/e = 65

The presence of these ions strongly suggests the presence of a benzyl group ($C_6H_5CH_2$–) in the compound being analyzed.

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56.15 Fragmentation patterns of aldehydes and ketones N10

N10 Refer to page T74.

Carbon-carbon bonds next to the carbonyl group of the molecular ion of an aldehyde or ketone break readily to generate RCO^+ ions. Consider the simplified mass spectrum of pentan-3-one (Fig. 56.31).

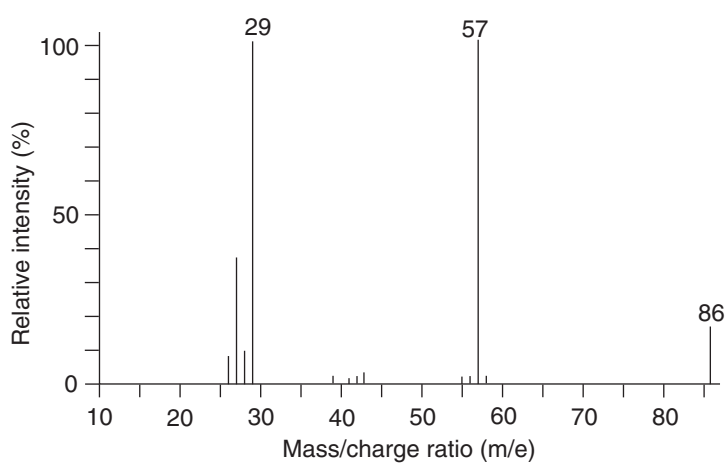


Fig. 56.31 The simplified mass spectrum of pentan-3-one

The peak at m/e = 57 corresponds to the $CH_3CH_2CO^+$ ion, which is produced by this fragmentation:

$$\underset{\substack{\text{molecular ion}\\ \text{m/e = 86}}}{\left[CH_3CH_2CO\{CH_2CH_3\right]^{+}\cdot} \longrightarrow \underset{\text{m/e = 57}}{CH_3CH_2CO^+} + \cdot CH_2CH_3$$

The peak at m/e = 29 corresponds to the ethyl ion — which could be formed by splitting the molecular ion at either side of the C=O group.

Typical fragment ions produced in the mass spectrometer are shown in Table 56.4.

Table 56.4 Typical fragment ions produced in the mass spectrometer

Ion	Mass/charge ratio (m/e)	Ion	Mass/charge ratio (m/e)
CH_3^+	15	$C_4H_9^+$	57
$C_2H_5^+$	29	$CH_3CH_2CO^+$	57
CH_3CO^+	43	$C_6H_5^+$	77
$C_3H_7^+$	43	$C_6H_5CH_2^+$	91

AL 2008 Paper 2 Q6(a)

DSE 2012 Paper 2 Q3(a)(iv)

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56.16 Differences in mass/charge ratio between peaks

Apart from deciding the ions responsible for the major peaks, the other important thing to look for is the differences in mass/charge ratio between the peaks. A difference of 15 almost certainly shows the loss of a methyl radical, and so it is likely that the original molecule contains a $-CH_3$ group.

Table 56.5 lists some characteristic differences in the mass/charge ratio between peaks that give clues to groups that may be present in the original molecule.

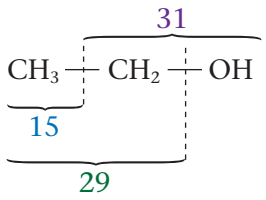
Table 56.5 Some common differences in mass/charge ratio due to the loss of fragments

Difference in mass/charge ratio	Commonly lost fragment	Group that may be present in the original molecule
$M^{+\cdot} - 15$	methyl radical	$-CH_3$
AL 2009 Paper 2 Q5(c) $M^{+\cdot} - 17$	hydroxyl radical	$-OH$
$M^{+\cdot} - 29$	ethyl radical	$-C_2H_5$

56.17 Interpreting mass spectra

Mass spectrum of ethanol

Fig. 56.32 shows the mass spectrum of ethanol (relative molecular mass = 46.0).



m/e	Ion
46	$[CH_3CH_2OH]^{+\cdot}$ molecular ion peak
31	CH_2OH^+
29	$CH_3CH_2^+$
15	CH_3^+

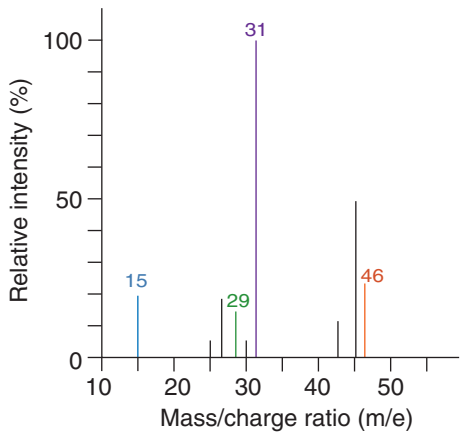


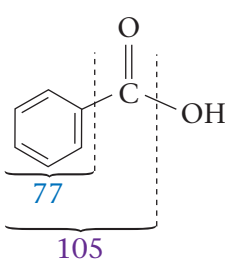
Fig. 56.32 The mass spectrum of ethanol

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The base peak is at m/e = 31. The difference between 46 and 31 is 15, so it is probable that a methyl radical has been lost.

Mass spectrum of benzoic acid

Fig. 56.33 shows the mass spectrum of benzoic acid, C_6H_5COOH (relative molecular mass = 122.0).



m/e	Ion
122	$[C_6H_5COOH]^{+\cdot}$
105	$C_6H_5CO^+$
77	$C_6H_5^+$

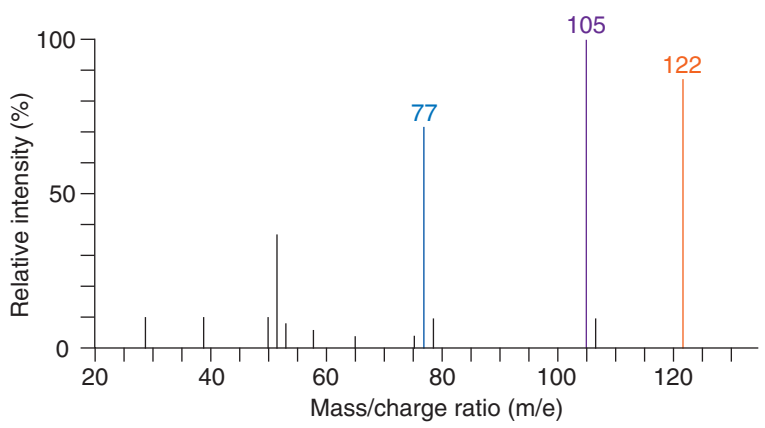


Fig. 56.33 The mass spectrum of benzoic acid

AL 2007 Paper 2 Q7(c)(ii)

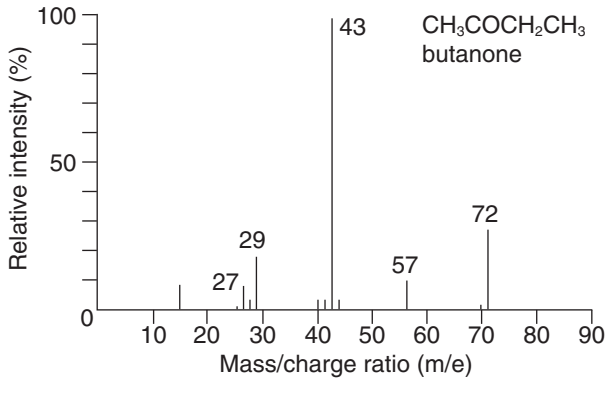
The base peak is at m/e = 105. The difference between 122 and 105 is 17, so it is probable that a hydroxyl radical has been lost.

A peak at m/e = 77 is frequently present in compounds with a benzene ring.

Checkpoint

Refer to page T75.

1 The mass spectrum of butanone is shown below.



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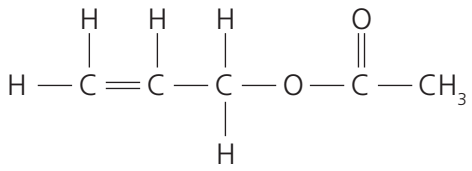
a) Identify the molecular ion peak.

b) Identify the base peak and suggest a possible ion responsible for the peak.

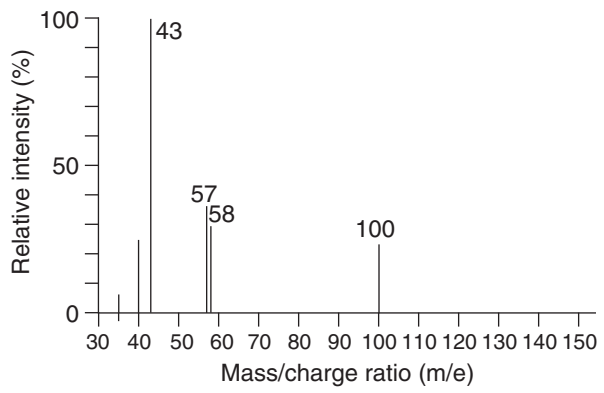
c) Write an equation to show the fragmentation that occurs when the molecular ion is converted into the ion responsible for the base peak.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

2 Consider an ester with the following structure:



The mass spectrum of the ester is shown below:



a) Identify the ions responsible for the peaks at m/e = 100, 57 and 43.

b) Suggest a fragment that is lost when the ion at m/e = 100 is converted into the ion at m/e = 43.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

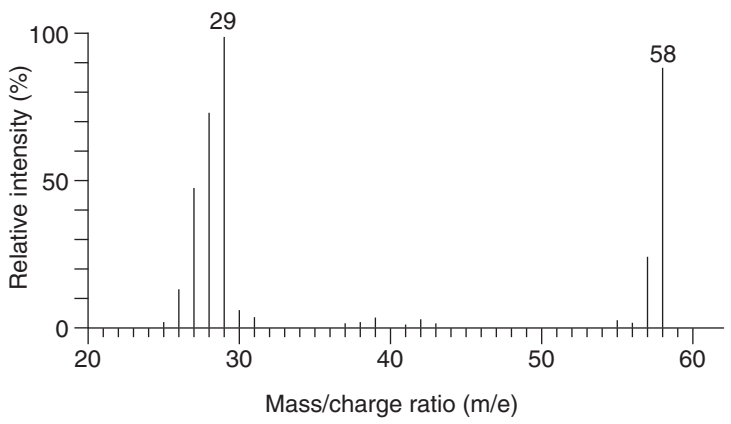
Study the following examples to see how we can use mass spectrometry to determine the structures of compounds.

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Example

56.4 Identifying a compound using its mass spectrum

The simplified mass spectra of compound A is shown below. The compound is known to be either propanal or propanone.



a) Write the structural formulae of propanal and propanone.

b) Identify the molecular ion peak in the spectrum.

c) Suggest whether compound A is propanal or propanone. Explain your answer.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

Solution

a) CH_3CH_2CHO CH_3COCH_3
propanal propanone

b) The peak at m/e = 58 is the molecular ion peak.

c) If compound A is propanone, it likely splits to generate the following two ions:

- CH_3^+
- CH_3CO^+

These would give key peaks at m/e = 15 and 43. These two peaks are not found in the spectrum. Thus compound A is not propanone.

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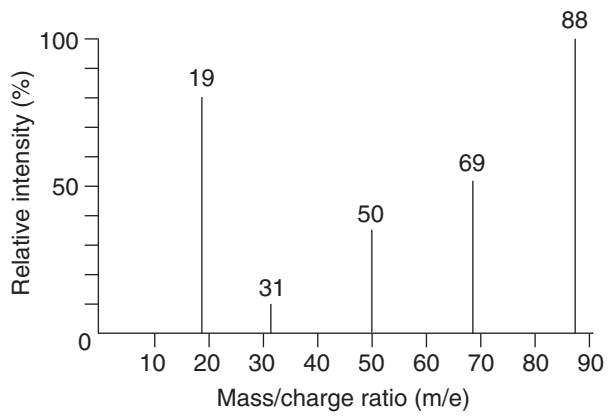
The peak at m/e = 29 corresponds to the CHO^+ ion, which is produced by this fragmentation:

$$\underset{\substack{\text{molecular ion of propanal}\\ \text{m/e = 58}}}{\left[CH_3CH_2\{CHO\right]^+\cdot} \longrightarrow \underset{\text{m/e = 29}}{CHO^+} + \cdot CH_2CH_3$$

This evidence suggests that compound A is propanal.

56.5 Deducing the structure of a compound from its mass spectrum

Elements X and Y form a compound with the molecular formula of XY_3. The mass spectrum of the compound is shown.



a) Identify the molecular ion peak.

b) Peaks at m/e = 69 and 50 are due to ions formed from the loss of one and two atoms of Y from the molecular ion respectively.

Which ion is probably responsible for the peak at m/e = 31?

c) Suggest the identity of compound XY_3.

AL 2002 Paper 2 Q1(b)

Solution

a) The peak at m/e = 88 is the molecular ion peak.

b) The loss of an atom of Y from the molecular ion produces a peak at m/e = 69 while the loss of two atoms of Y produces a peak at m/e = 50. It can be deduced that the relative atomic mass of Y is 19.

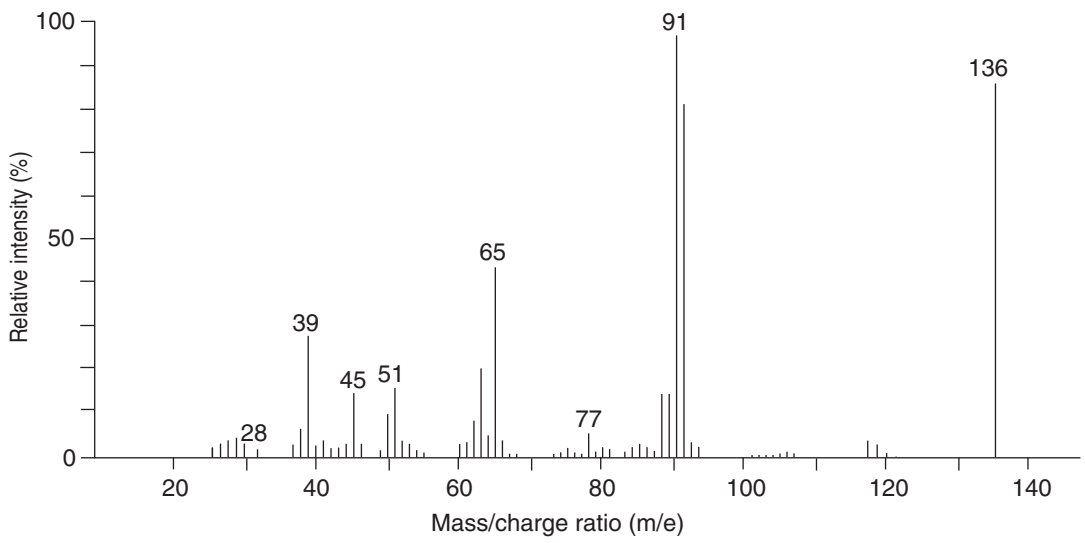
The loss of three atoms of Y from the molecular ion produces a peak at m/e = 31. X^+ is formed here and hence the relative atomic mass of X is 31.

c) The compound XY_3 is phosphorus trifluoride.

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56.6 Deducing the structure of a compound from its mass spectrum

The mass spectrum of a compound X, $C_8H_8O_2$ (relative molecular mass = 136.0), is shown below.



a) Identify the molecular ion peak.

b) Suggest the ion responsible for the peak at m/e = 91.

c) Suggest a possible structure of compound X.

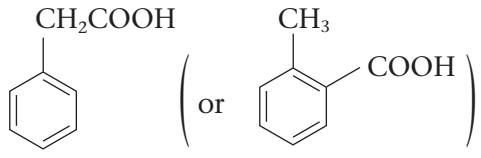
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

Solution

a) The peak at m/e = 136 is the molecular peak.

b) The $C_6H_5CH_2^+$ ion is probably responsible for the peak at m/e = 91.

c) A possible structure of compound X is:



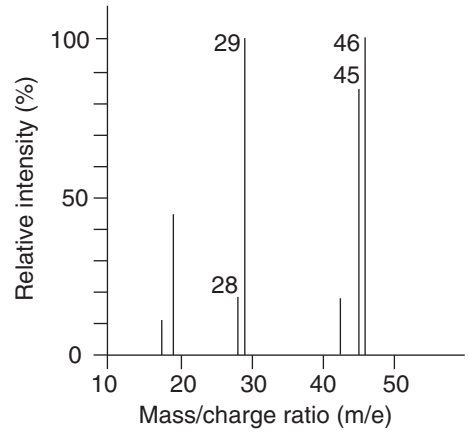
Unit 56(E).indd 192 2015/6/30 11:41:44 AM

Checkpoint

Refer to page T76.

1 While running a new reaction, a chemist noticed the evolution of a gas. A sample of this gas gave a mass spectrum in which the molecular ion (m/e = 44) was the base peak. The only other significant peaks were observed at m/e = 28 and m/e = 16. What is this gas?

2 A compound has the mass spectrum shown below. It has a molecular formula of CH_2O_2 (relative molecular mass = 46.0).



a) What fragment could be lost from the molecular ion to form a peak at m/e = 29?

b) Which ions are probably responsible for the peaks at m/e = 28, 29 and 45 respectively?

c) Suggest a possible structure of this compound.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

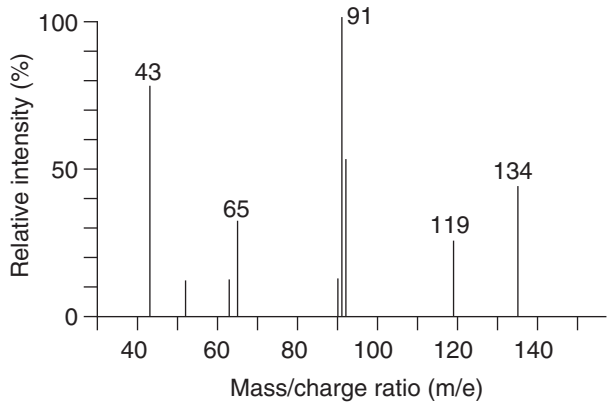
N11 Refer to page T77.

Example

56.7 Deducing the structure of a compound from its infrared and mass spectra N11

The molecular formula of an aromatic compound X is $C_9H_{10}O$. X does not react with aqueous bromine. It has an infrared absorption at about 1 700 cm^{-1}. The mass spectrum of X is shown below.

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a) Refer to the mass spectrum of compound X.

i) Identify the molecular ion peak. DSE 2014 Paper 2 Q3(c)(iii)

ii) Which fragment may be lost from the molecular ion to form a peak at m/e = 119?

iii) Which ions are probably responsible for the peaks at m/e = 91 and 43? DSE 2014 Paper 2 Q3(c)(iii)

b) Deduce a possible structure of compound X. DSE 2014 Paper 2 Q3(c)(iv)

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

Solution

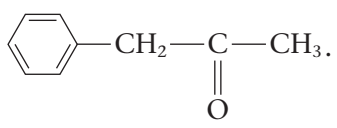
a) i) The peak at m/e = 134 is the molecular ion peak.

ii) The difference between 134 and 119 is 15, so it is probable that a methyl radical has been lost.

iii)

m/e	Ion
91	$C_6H_5CH_2^+$
43	CH_3CO^+

b)
- The infrared absorption at 1 700 cm^{-1} shows the presence of a C=O group.
- In the mass spectrum, the peak at m/e = 119 suggests the loss of a methyl radical from the molecular ion. Thus, a methyl group is likely to be present in X.
- The base peak at m/e = 91 is probably due to the $C_6H_5CH_2^+$ ion. This suggests the presence of a $C_6H_5CH_2$– group in X.
- The peak at m/e = 43 is probably due to the CH_3CO^+ ion. This suggests that X may be a methyl ketone.

Therefore a possible structure of X is  .

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Checkpoint

Refer to page T78.

1 Consider the following two compounds:

CH_3COOCH_3 (A) and $CH_3CH_2CH_2OH$ (B)

a) Describe how you could distinguish between the two compounds using a chemical test. State also the expected observations.

b) i) State how compounds A and B could be distinguished by their infrared spectra, without using the fingerprint region.

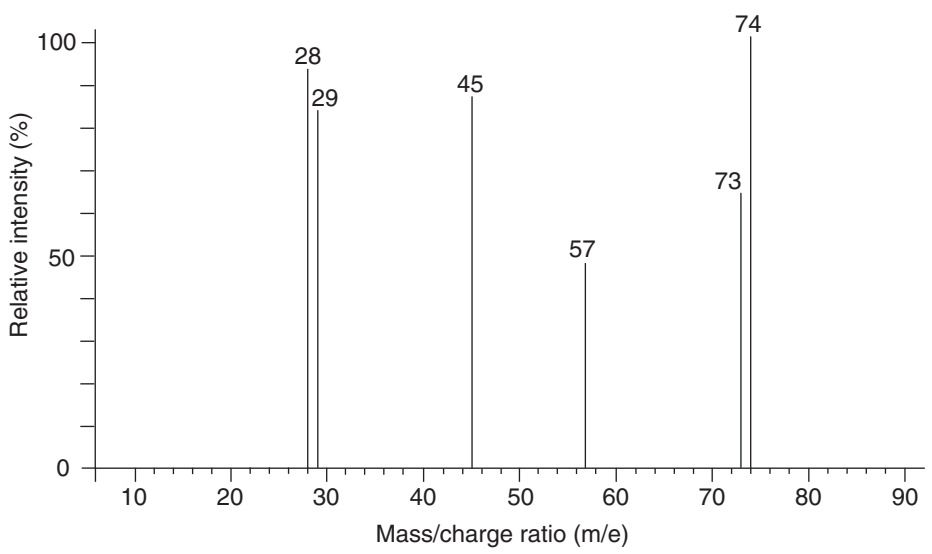
ii) Explain how the fingerprint region is used to identify a compound.

c) i) Suggest the structure of the fragment responsible for the base peak in the mass spectrum of compound A and state its m/e value.

ii) Write an equation showing the formation of this fragment from the molecular ion.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

2 The molecular formula of compound X is $C_3H_6O_2$. X is miscible with water. The simplified mass spectrum of X is shown below:



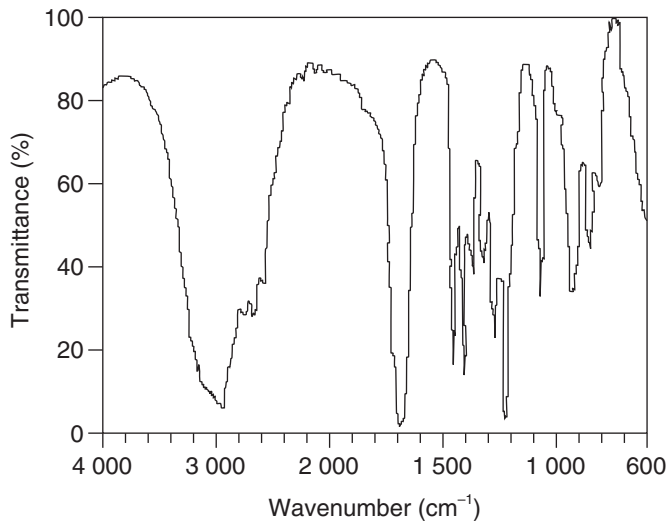
a) What fragment may be lost from the molecular ion to form a peak at m/e = 57?

b) What fragment may be lost from the molecular ion to form a peak at m/e = 45?

c) Which ion is probably responsible for the peak at m/e = 29?

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d) The infrared spectrum of X contains a broad absorption band at 2 500 cm^{-1} – 3 300 cm^{-1}. Identify the bond responsible for this absorption.



e) Suggest a possible structure of X.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

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Summary

1 Instrumental analytical methods have advantages over the more traditional gravimetric and volumetric analyses, including

- speed (therefore allowing for the testing of more samples);
- increased level of accuracy;
- less room for human error;
- sensitivity;
- non-destructiveness.

2 a) Colorimetry determines the concentration of a substance based on its ability to absorb certain parts of radiation of the visible spectrum.

b) The concentration of a species being analyzed in a sample solution can be determined from a calibration curve.

3 a) Infrared (IR) spectroscopy is the absorption measurement of infrared radiation of different wavenumbers by a sample.

b) The main goal of infrared spectroscopy is to determine the functional groups present in carbon compounds as different covalent bonds absorb infrared radiation of characteristic wavenumbers.

c) You need to refer to characteristic infrared absorption wavenumber ranges for some covalent bonds in Table 56.3 when identifying bonds or functional groups in carbon compounds.

4 a) Mass spectrometry is a form of analysis that separates and identifies substances on the basis of mass-to-charge ratios of positive ions formed from substances when they are bombarded by high energy electrons in a vacuum.

b) Mass spectra can be used to measure relative atomic masses and relative molecular masses.

c) Structural information of a compound can be determined from fragmentation patterns of the molecular ion. The following table shows typical fragment ions produced in a mass spectrometer.

Ion	Mass/charge ratio (m/e)	Ion	Mass/charge ratio (m/e)
CH_3^+	15	$C_4H_9^+$	57
$C_2H_5^+$	29	$CH_3CH_2CO^+$	57
CH_3CO^+	43	$C_6H_5^+$	77
$C_3H_7^+$	43	$C_6H_5CH_2^+$	91

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Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Analytical methods can be classified according to the property that is observed in the measurement process. Complete the table.

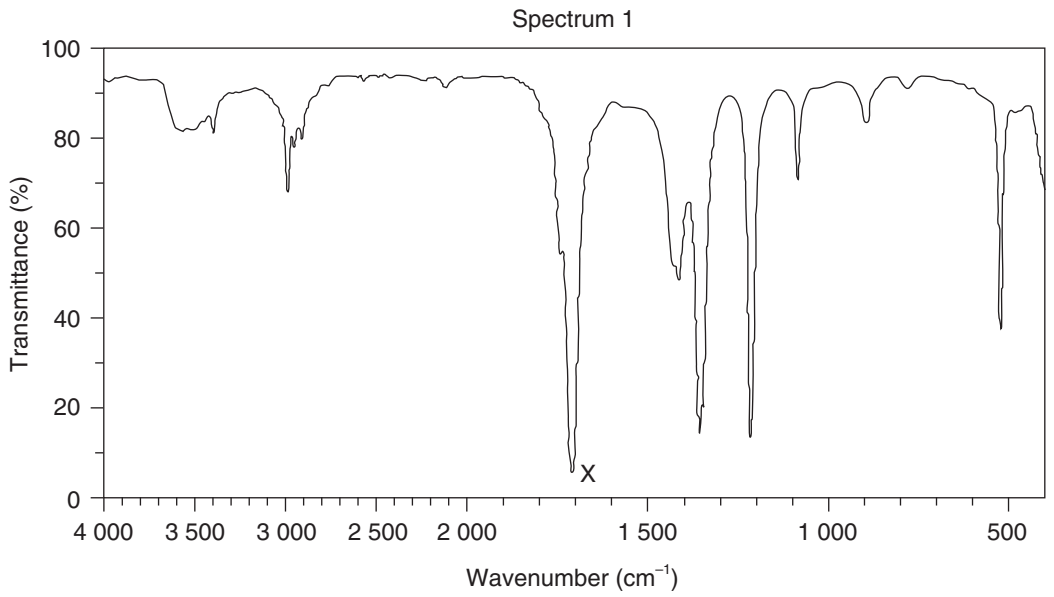
Physical property measured	Analytical method based on measurement of the property
Mass	gravimetric analysis
Volume	volumetric analysis
Absorption of electromagnetic radiation	colorimetry / infrared spectroscopy
Mass-to-charge ratio	mass spectrometry

2 The infrared spectra, spectrum 1 and spectrum 2, are those of propan-2-ol and propanone.
Refer to page T79.

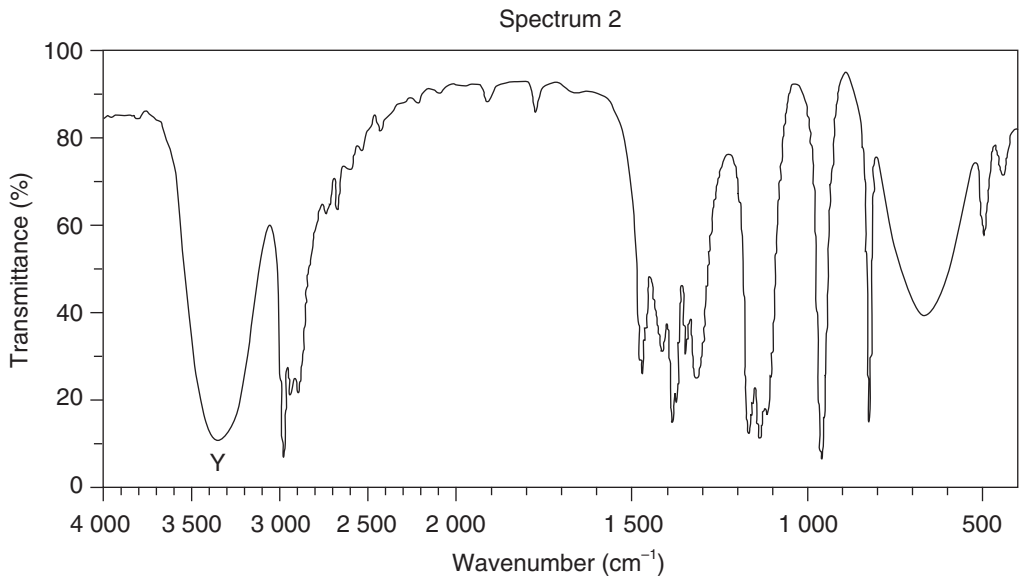
a) Draw the structural formulae of propan-2-ol and propanone.

b) Identify the functional groups which cause the absorptions labelled X and Y.

c) Decide which spectrum represents propan-2-ol and which represents propanone.



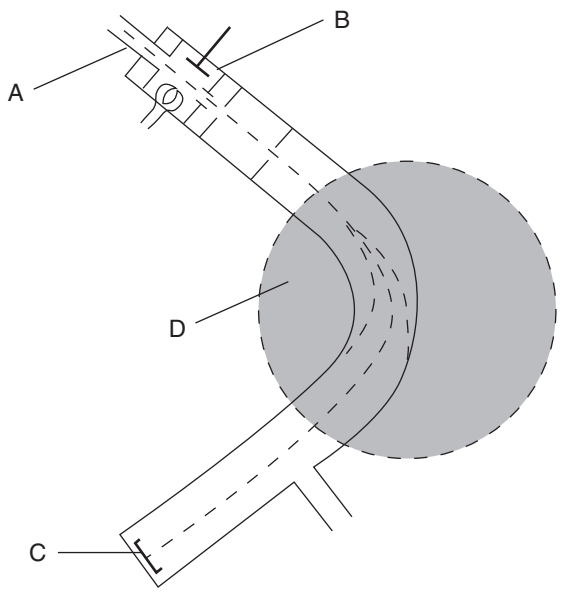
Unit 56(E).indd 198 2015/6/30 11:41:51 AM



3 a) Choose FOUR of the following terms to label the simplified diagram of a mass spectrometer below.

ionization chamber, sample inlet, ion detector, magnetic field, vacuum pump, accelerating electric field

A — sample inlet
B — ionization chamber
C — ion detector
D — magnetic field



b) What is meant by the following terms used in mass spectrometry?

i) Molecular ion peak

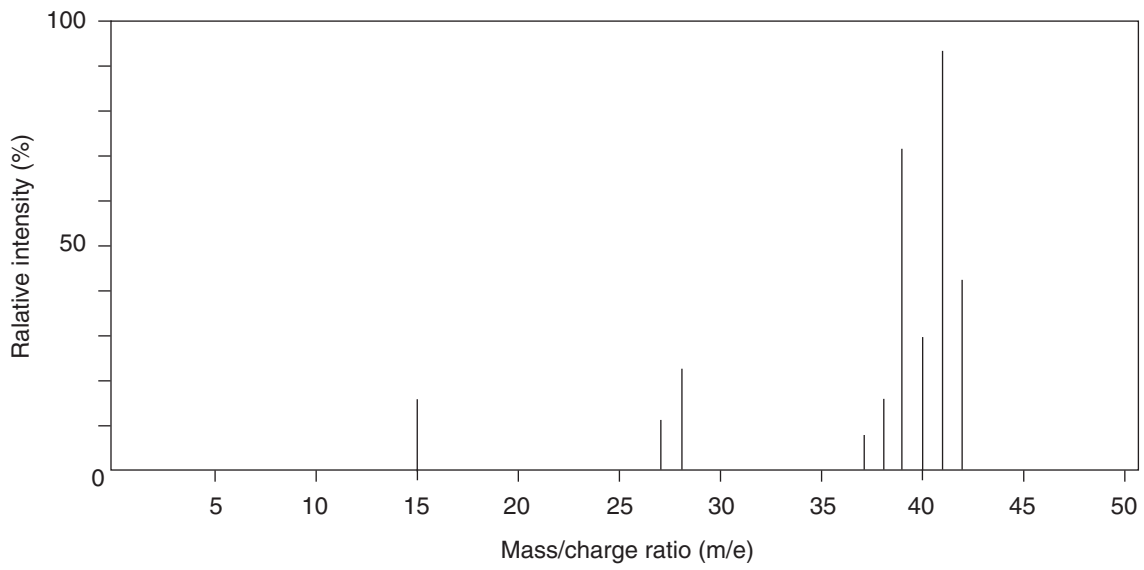
The peak at the m/e which equals the relative molecular mass of the sample

ii) Base peak

The strongest detector signal which is set to 100%

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4 The major peaks in the mass spectrum of a hydrocarbon A are shown below.



a) Deduce the molecular formula of hydrocarbon A.

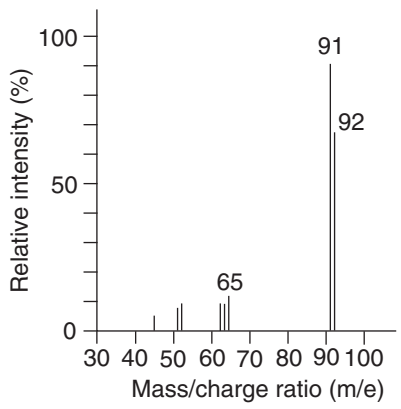
b) Draw the structural formula of hydrocarbon A.

(Relative atomic masses: H = 1.0, C = 12.0)

a) Let C_xH_y be the molecular formula of A.
The peak at m/e = 42 is the molecular ion peak.
Relative molecular mass of A = 42.0 = 12.0x + 1.0y
x = 3 and y = 6
∴ the molecular formula of A is C_3H_6.

b) The difference between 42 and 27 is 15, so it is probable that a methyl radical is lost from the molecular ion, producing the peak at m/e = 27.
The structural formula of hydrocarbon A is $CH_3CH{=}CH_2$.

5 The mass spectrum of methylbenzene is shown below.
Refer to page T79.



a) Identify the species responsible for the peak at m/e = 91 and also that responsible for the peak at m/e = 65.

b) Write an equation for the formation of the species responsible for the peak at m/e = 65.

(Relative atomic masses: H = 1.0, C = 12.0)

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Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 In an experiment to determine the concentration of H_2O_2 in a rainwater sample, 5.00 cm^3 of the sample were mixed with an excess of a certain transition metal complex solution, giving a coloured mixture. The absorbance of the mixture was measured by a colorimeter and was found to be 0.278. When 5.00 cm^3 of 1.55 x 10^{-6} mol dm^{-3} H_2O_2 solution were treated in the same way, an absorbance of 0.410 was recorded.

Refer to page T79.

a) Draw a labelled diagram of a colorimeter and explain its operating principle.

b) Calculate the concentration of H_2O_2 in the rainwater sample, assuming that the concentration is directly proportional to the absorbance. 1.05 x 10^{-6} mol dm^{-3}

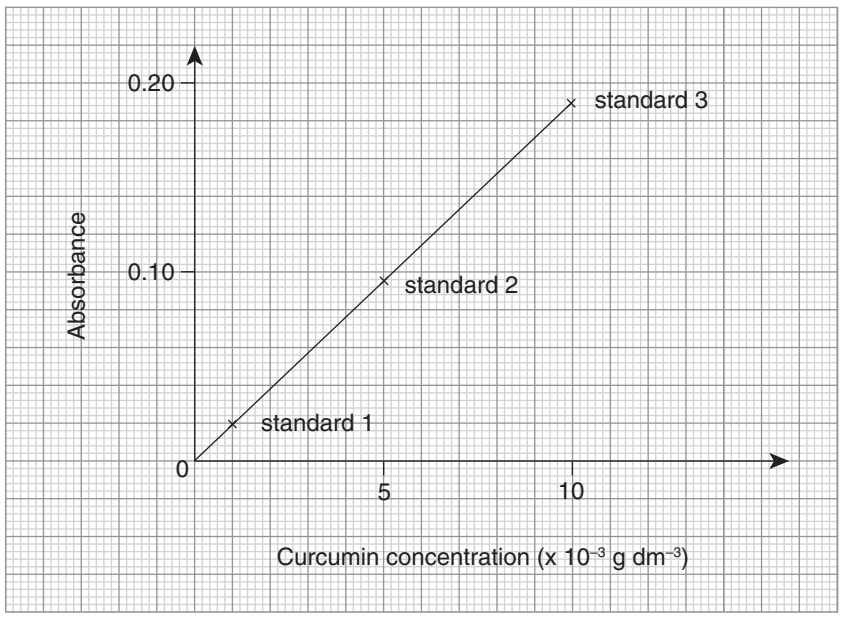
7 The yellow chemical curcumin is used in food dyes.

Refer to page T80.

In one analysis, a stock solution of curcumin was prepared by dissolving 0.100 g of curcumin in 250.0 cm^3 of water. The stock solution was then diluted to make the following three standard solutions.

Standard solution	Curcumin concentration (g dm^{-3})
Standard 1	1.00 x 10^{-3}
Standard 2	5.00 x 10^{-3}
Standard 3	1.00 x 10^{-2}

The absorbance of each standard is measured using a colorimeter and the following calibration curve is obtained.



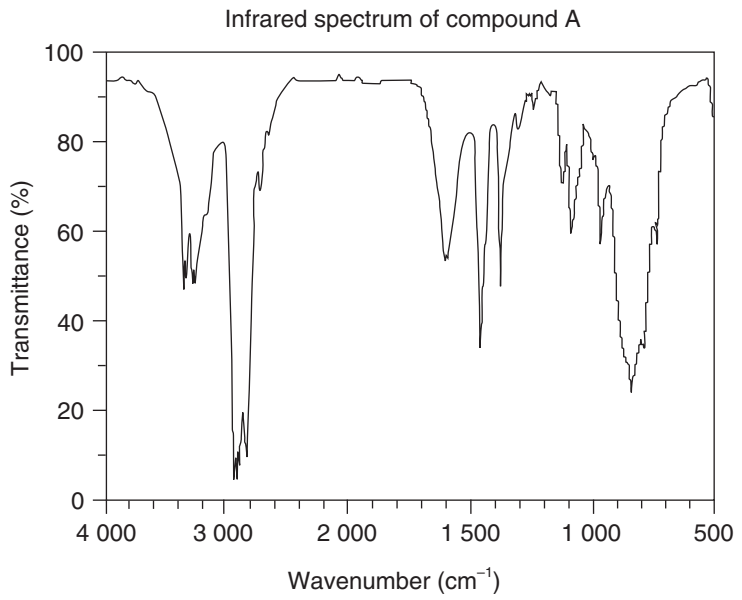
Unit 56(E).indd 201 2015/6/30 11:41:54 AM

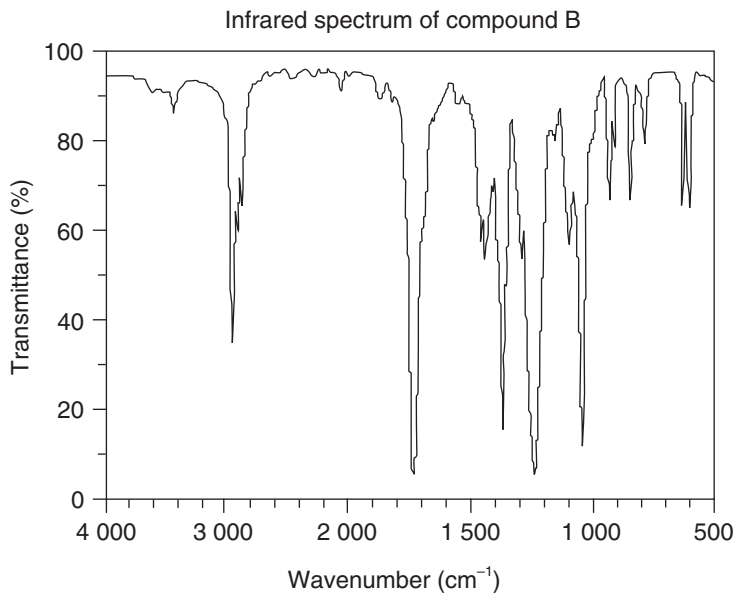
a) What volume of water was added to 10.0 cm^3 of the stock solution to make standard 3? 390 cm^3

b) The curcumin in a 24.5 g food sample was extracted into solution. The solution was made up to 250.0 cm^3. The absorbance of the solution was found to be 0.170.

Calculate the curcumin content in the food sample in milligram per gram of the sample.
0.0910 mg per gram of food sample

8 The infrared spectra of two compounds, A and B, are shown below. The compounds are known to be ethyl ethanoate and butan-1-amine, though not necessarily in that order.
Refer to page T80.





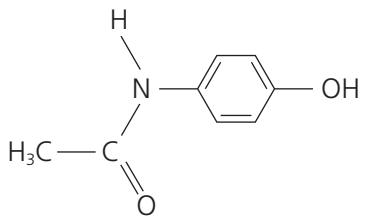
a) Write the structural formulae of ethyl ethanoate and butan-1-amine.

b) Identify compounds A and B.

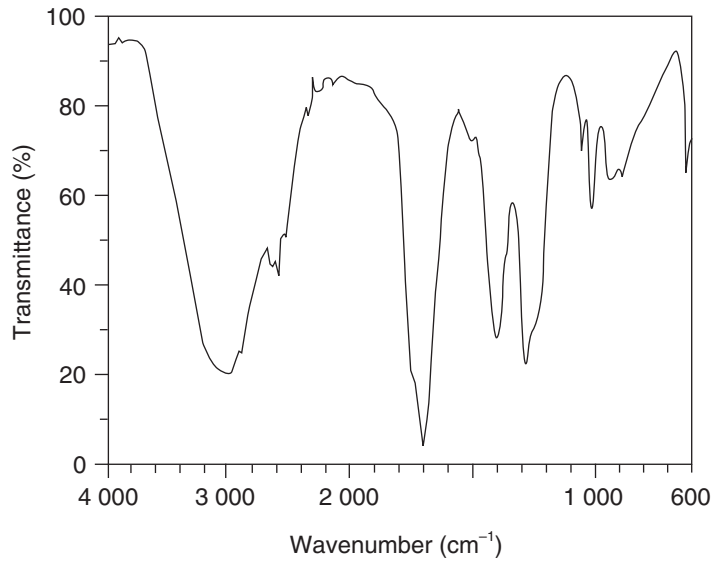
Exercise

Unit 56(E).indd 202 2015/6/30 11:41:55 AM

9 Acetaminophen is a medicine that reduces the symptoms of a fever.
Refer to page T81.



A stored sample of acetaminophen is damp and some hydrolysis has taken place to give the impurities. The infrared spectrum of ONE of the impurities, X, is shown below.

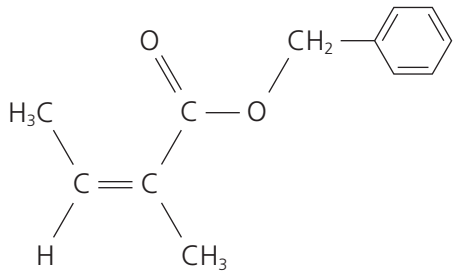


a) Identify the functional group in X. Give your reasoning by identifying the major absorptions in the spectrum and the bond to which each corresponds.

b) The molar mass of X is 60.0 g mol^{-1}. Draw the structural formula of X.

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0)

10 Compound A is used to add the flavour of mushrooms to foods.
Refer to page T81.



compound A

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a) i) Apart from the benzene ring, name the TWO functional groups in compound A.

ii) Draw the skeletal formula of compound A.

iii) Deduce the molecular formula of compound A.

b) Compound B is a stereoisomer of compound A. Explain what is meant by the term 'stereoisomerism'. Use compounds A and B to illustrate your answer.

c) If the food is cooked for a long time, naturally occurring acids catalyze the hydrolysis of compound A.

Draw structures to show the TWO carbon compounds formed by the acid hydrolysis of compound A.

d) The hydrolysis of compound A can be monitored by sampling the mixture at regular intervals, separating the components, and recording their infrared spectra.

i) State TWO absorptions that would be expected in the infrared spectrum of compound A, and identify the parts of the molecule responsible for each.

ii) Suggest a wavenumber range within the spectrum that could be used to clearly distinguish compound A from the products formed by the hydrolysis reaction. Explain your answer.

(OCR Advanced Level GCE, Chains, Rings and Spectroscopy, Jan. 2006)

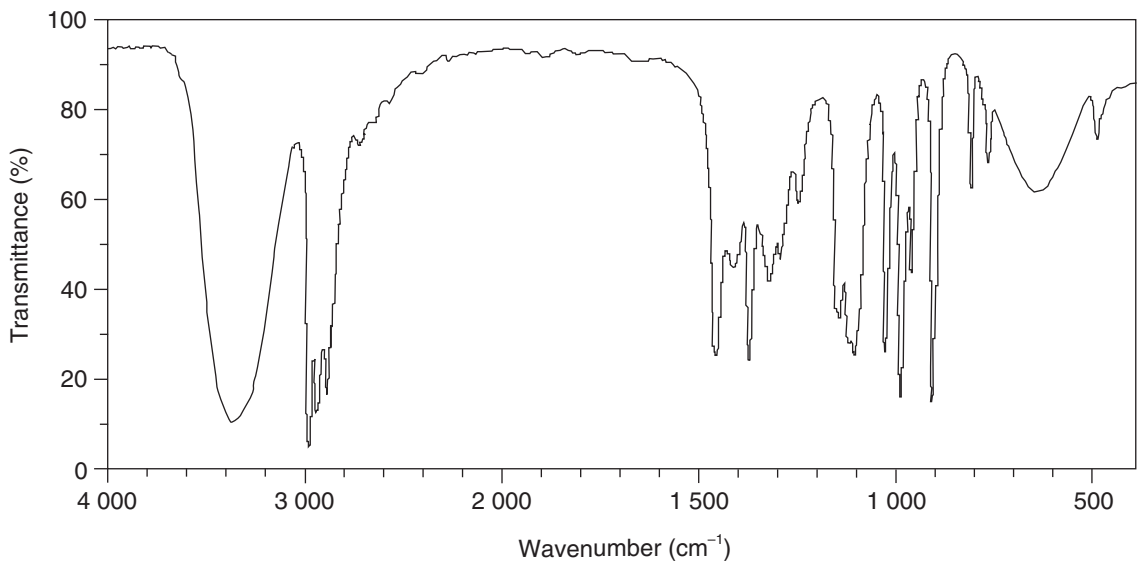
Exercise

11 Compound X has the molecular formula $C_4H_{10}O$.

Refer to page T82.

a) Draw SIX possible structures of compound X.

b) The infrared spectrum of compound X is shown below and X can exist as a pair of enantiomers. Suggest a possible structure of X and explain briefly your reasoning.



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12 Deduce the structure of compound D on the basis of the information given below:

(1) Elemental analysis data show that D has the following composition by mass: C 40.0%, H 6.7% and O 53.3%.

(2) The relative molecular mass of D is estimated to be in the range of 172 to 182.

(3) The infrared spectrum of D shows, apart from the absorption of C–H stretching near 2 900 cm^{-1}, a strong and broad absorption around 3 400 cm^{-1}, and no appreciable absorption around 1 700 cm^{-1}.

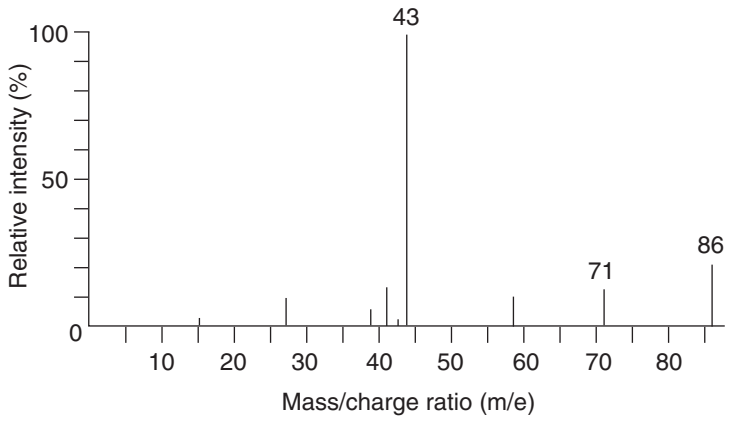
(4) All carbon atoms of D have the same bonding environment.

(5) D is highly soluble in water, and the solution does not decolorize bromine water.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

(HKALE 2006)

13 The simplified mass spectrum of a compound is shown below. The compound is known to be either pentan-2-one or pentan-3-one.



a) Write the structural formulae of pentan-2-one and pentan-3-one.

b) Identify the molecular ion peak in the spectrum.

c) Identify the species responsible for the peak at m/e = 71 and that responsible for the peak at m/e = 43.

d) Suggest whether the compound is pentan-2-one or pentan-3-one. Explain your answer.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

a) Pentan-2-one $CH_3CH_2CH_2COCH_3$
Pentan-3-one $CH_3CH_2COCH_2CH_3$

b) The peak at m/e = 86

c) The difference between 86 and 71 is 15, so it is probable that a methyl radial is lost from the molecular ion, producing the peak at m/e = 71.
The CH_3CO^+ ion is responsible for the peak at m/e = 43.

d) The compound is pentan-2-one.
The compound is not pentan-3-one because its mass spectrum should have a peak at m/e = 57 due to the $CH_3CH_2CO^+$ ion.
The peak at m/e = 71 is produced by this fragmentation:
$[CH_3CH_2CH_2CO\{CH_3]^{\ddagger\bullet} \longrightarrow CH_3CH_2CH_2CO^+ + \bullet CH_3$
molecular ion
m/e = 86
The peak at m/e = 43 is produced by this fragmentation:
$[CH_3CH_2CH_2\{COCH_3]^{\ddagger\bullet} \longrightarrow CH_3CO^+ + \bullet CH_2CH_2CH_3$

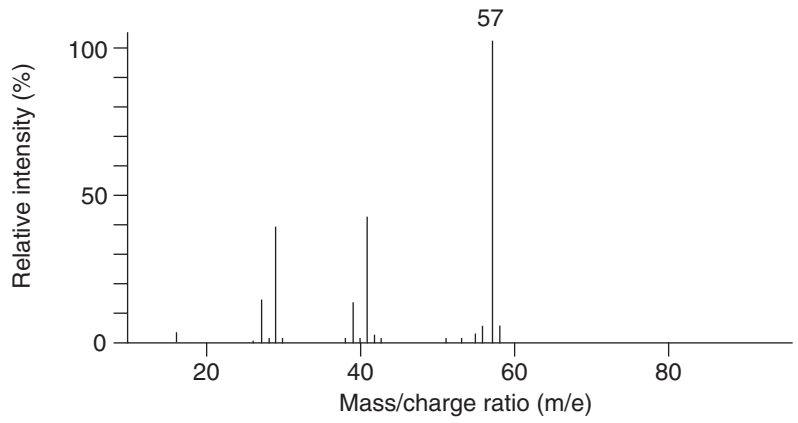
Unit 56(E).indd 205 2015/6/30 11:41:59 AM

14 Alkane A has a molecular formula C_5H_{12}.

Refer to page T82.

a) Draw the structures of all isomers of A.

b) The mass spectrum of A is shown below. By considering the intensity of the labelled peak, suggest the most possible structure of A. Explain your answer.

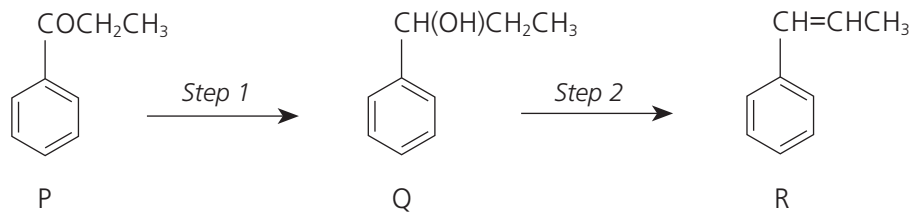


(Relative atomic masses: H = 1.0, C = 12.0)

Exercise

15 Compound P can be converted to Q and then to R as shown below:

Refer to page T83.



a) The mass spectrum of P contains a molecular ion peak at m/e = 134 and major fragmentation peaks at m/e = 105 and 77.

i) Identify the species responsible for the peak at m/e = 105 and also that responsible for the peak at m/e = 77.

ii) Write an equation for the formation, from the molecular ion, of the species responsible for the peak at m/e = 105.

b) i) Suggest suitable reagents and reaction conditions for the reaction in *Step 1*.

ii) Q can exist in isomeric forms. Draw a suitable representation for each of these isomers.

c) i) Suggest suitable reagents and reaction conditions for the reaction in *Step 2*.

ii) Name the type of stereoisomerism shown by the product of this reaction.

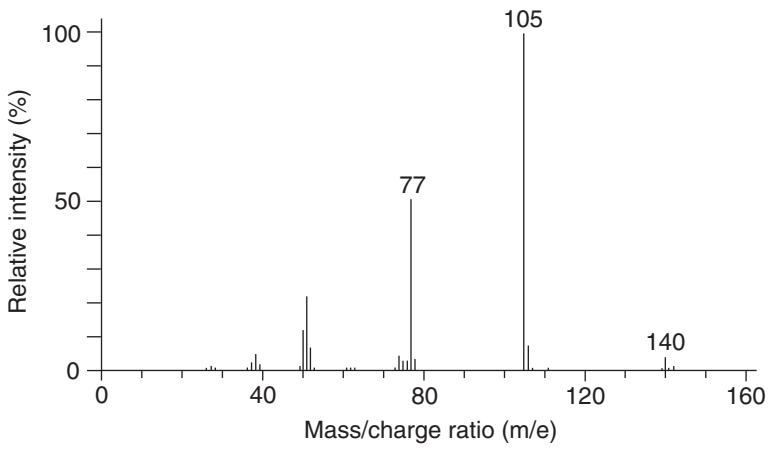
iii) State what is required in the structure of molecules to allow them to show the type of stereoisomerism suggested in (ii).

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16 Compound M has the following composition by mass:

C 59.8% H 3.6% Cl 25.2% O 11.4%

M displays the mass spectrum as shown below:



a) Calculate the empirical formula of M.

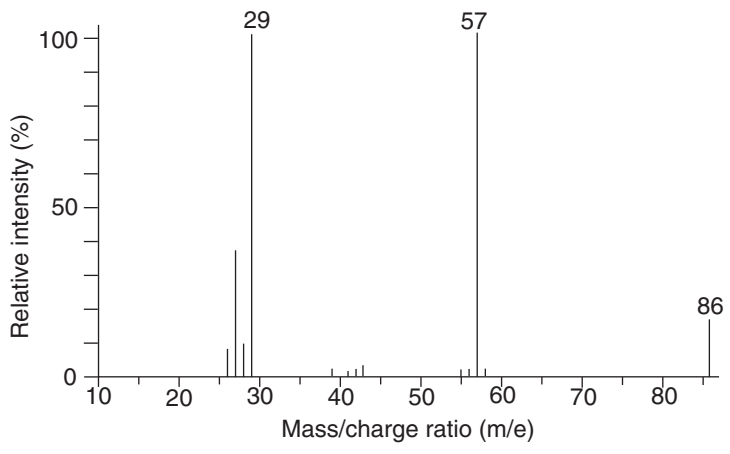
b) Deduce the structure of M.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Cl = 35.5)

(HKALE 2007)

17 Compound A has the following composition by mass: 69.8% of C, 11.6% of H, and 18.6 % of O. The mass spectrum of A is shown below.

Refer to page T84.



a) Determine the empirical formula of compound A. $C_5H_{10}O$

b) Using the information in the mass spectrum, determine the molecular formula of compound A. $C_5H_{10}O$

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c) Compound A reacts with 2,4-dinitrophenylhydrazine to give an orange precipitate, and give no observable change when treated with acidified aqueous solution of potassium dichromate.

Draw the structural formulae of all possible isomers of this compound.

d) Study the above mass spectrum.

i) Identify the fragment ion responsible for the peak at m/e = 29 and that responsible for the peak at m/e = 57.

ii) Propose a structure of compound A.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

18 Two isomeric alcohols A and B have the formula $C_4H_{10}O$.

Refer to page T85.

The mass spectrum of A shows a major peak at m/e (M – 45) but no peak at m/e (M – 43).

The mass spectrum of B shows major peaks at *both* m/e (M – 45) and m/e (M – 43).

Suggest a reason for these different fragmentation patterns, and hence the structural formulae for A and B.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

(OCR Advanced GCE, Methods of Analysis and Detection, Jan. 2005)

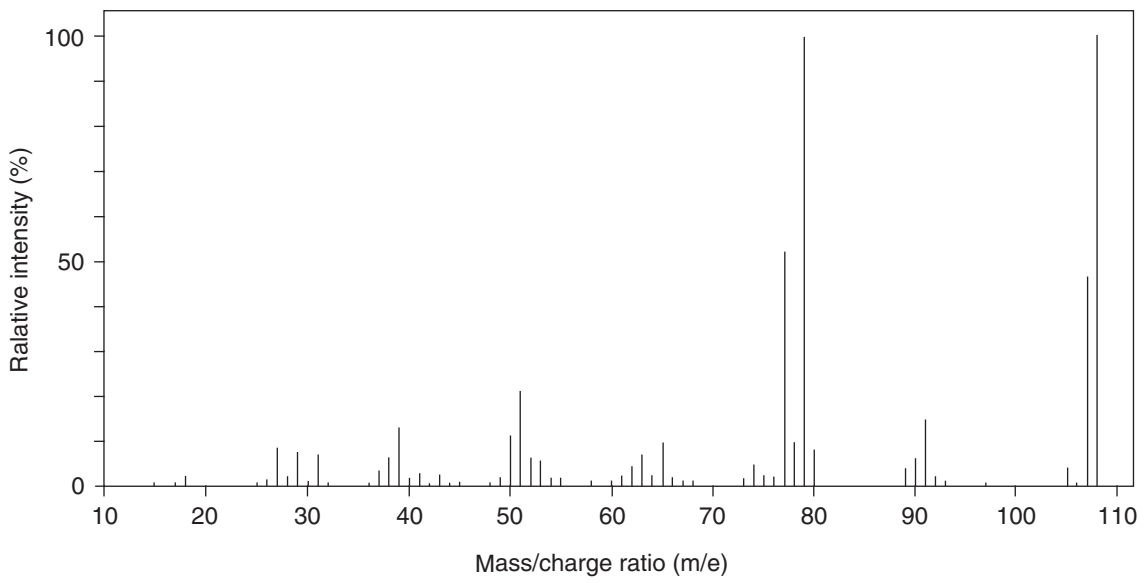
Exercise

19 Compound X has the following composition by mass:

Refer to page T86.

C 77.8% H 7.40% O 14.8%

a) The mass of spectrum of compound X is shown below.

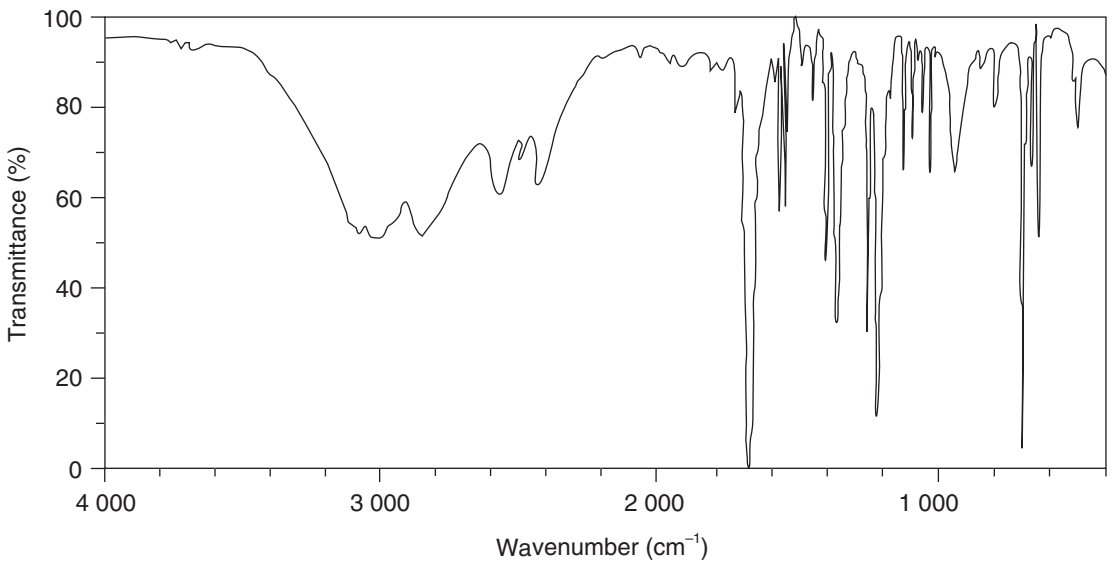


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i) What is the m/e of the molecular ion of compound X? 108

ii) What fragment ion is responsible for the peak at m/e = 77?

b) Compound X undergoes oxidation with acidified $Na_2Cr_2O_7(aq)$ to give compound Y. The infrared spectrum of Y is shown below.



Based on the above information, deduce the structures of compounds X and Y. Explain your reasoning.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

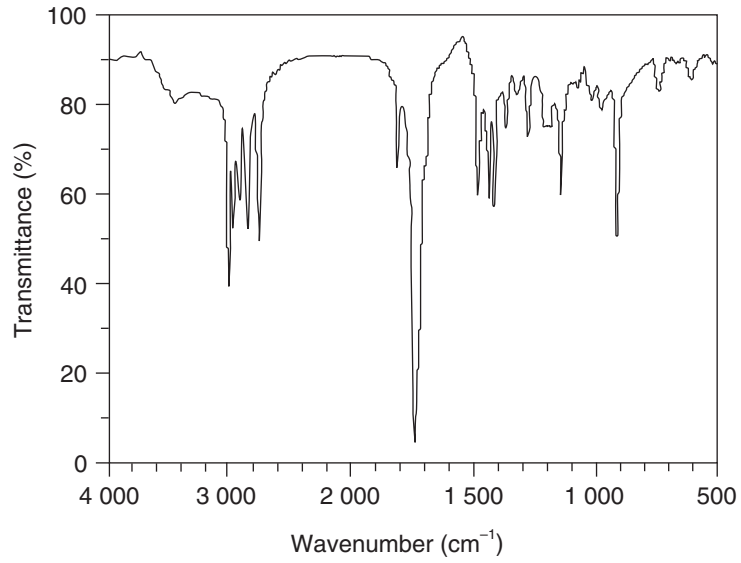
20 A carbon compound A, $C_xH_yO_z$, was found to contain 66.7% of carbon and 11.1% of hydrogen by mass. Compound A produced a strong infrared absorption at 1 720 cm^{-1} and its mass spectrum contained lines at m/e = 72 (the M peak), 57 and 43.
Refer to page T87.

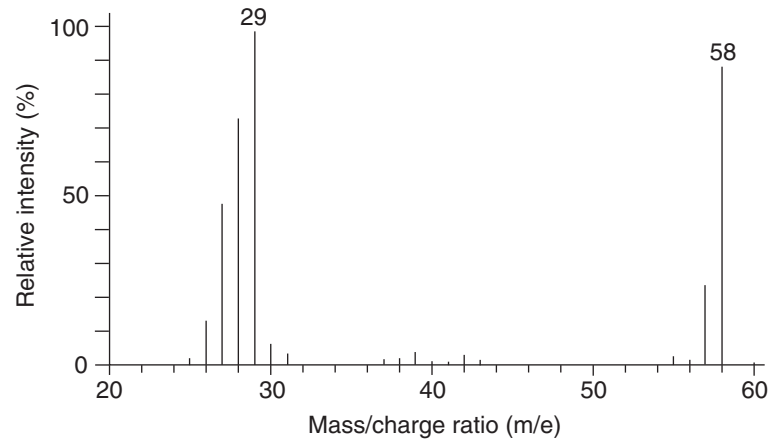
Use these data to suggest a structure of compound A, explaining your reasoning.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

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21 The infrared spectrum and mass spectrum of compound X are shown below. Deduce the structure of X. Explain your reasoning.





(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

The absorption at 1 680 – 1 750 cm^{-1} indicates the presence of a C=O bond.
The molecular ion peak of compound X is at m/e = 58. Thus, its relative molecular mass is 58.0.
Other than the oxygen atom (relative atomic mass = 16.0), there is probably 3 carbon atoms and 6 hydrogen atoms in compound X.
Hence compound X is probably an aldehyde or a ketone.
The peak at m/e = 29 in the mass spectrum is due to the $C_2H_5^+$ ion. Hence compound X probably contains an ethyl group.
Thus, the structure of compound X is CH_3CH_2CHO.

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